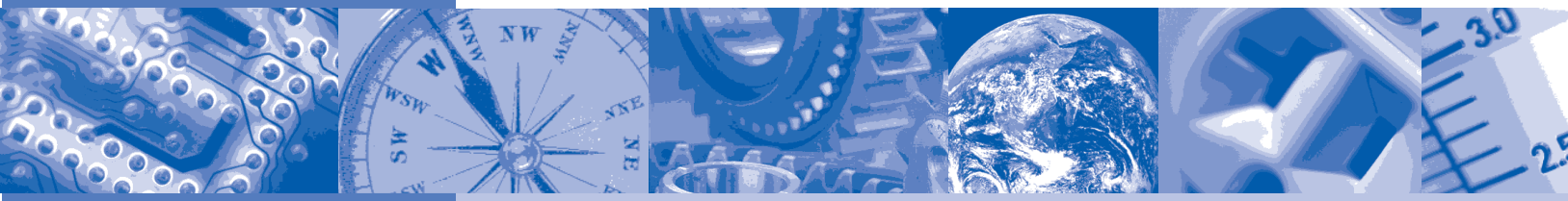
STANDARD

ISA-18.1-1979 (R2004)

Formerly ISA-18.1-1979 (R1992)



Annunciator Sequences and Specifications

Reaffirmed 25 February 2004

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ISA-18.1-1979 (R2004)
Annunciator Sequences and Specifications

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Preface

This preface is included for informational purposes and is not part of ISA-18.1-1979 (R2004).

This standard has been prepared as part of the service of ISA toward a goal of uniformity in the field of instrumentation. To be of real value, this document should not be static, but should be subject to periodic review. Toward this end, the Society welcomes all comments and criticisms and asks that they be addressed to the Secretary, Standards and Practices Board, ISA, 67 Alexander Drive, P.O. Box 12277, Research Triangle Park, North Carolina 27709, Telephone (919) 549-8411, e-mail: standards@isa.org.

Based on work started in 1955 by a survey committee titled Instrument Alarms and Interlocks, the 8D-RP18 Committee on Annunciator Systems of the Production Processes was formed in 1969. Tentative Recommended Practice ISA-RP18.1, titled Specifications and Guides for the Use of General Purpose Annunciators, was completed by that Committee in 1965.

The committee, reactivated as Committee SP18, Instrument Signals and Alarms, began revising ISA-RP18.1 in 1976 to reflect current industry practice for annunciators; in 1991, the Nuclear Power Plant Standards Committee, SP67, assumed responsibility for the reaffirmation of this standard.

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The information contained in the preface, footnotes, and appendices is included for information only and is not a part of the standard.

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This reaffirmed standard was approved for publication by the ISA Standards and Practices Board in 2004.

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1 Purpose

The purpose of this Standard is to establish uniform annunciator terminology, sequence designations, and sequence presentation and to assist in the preparation of annunciator specifications and documentation.

This Standard is intended to improve communications among those that specify, distribute, manufacture, or use annunciators.

2 Scope

This Standard is primarily for use with electrical annunciators that call attention to abnormal process conditions by the use of individual illuminated visual displays and audible devices. Annunciators can range from a single annunciator cabinet, to complex annunciator systems with many lamp cabinets and remote logic cabinets.

The sequence designations provided can be used to describe basic annunciator sequences and also many sequence variations. This Standard lists types of information that should be included in annunciator specifications and types of documents that should be provided by manufacturers; however, detailed design requirements and documentation formats are beyond the scope of this Standard.

3 Definition of terms

The following are terms and their definitions that have special meaning in relation to annunciators. Commonly used alternate terms are shown in parentheses. Defined terms used in other definitions are in italics to provide a cross-reference.

acknowledge: the *sequence action* that indicates recognition of a new *alarm*.

active alarm point: see *alarm point*.

alarm: 1. an abnormal *process condition*. 2. the *sequence state* when an abnormal *process condition* occurs. 3. a device that calls attention to the existence of an abnormal *process condition*. See *annunciator*. Types of *alarm* include:

momentary: an *alarm* that returns to normal before being acknowledged.

maintained: an *alarm* that returns to normal after being acknowledged.

alarm module (point or sequence module): a plug-in assembly containing the sequence logic circuit. Some *alarm* modules also contain *visual display* lamps or lamps and *windows*.

alarm point: the sequence logic circuit, *visual display*, auxiliary devices, and internal wiring related to one *visual display*. Types of *alarm point* include:

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active: an *alarm point* that is wired internally and completely equipped. The *window* is labeled to identify a specific monitored variable.

spare: an *alarm point* that is wired internally and completely equipped. The *window* is not labeled to identify a monitored variable.

future (blank): an *alarm point* that is wired internally and equipped except for the plug-in *alarm module.* The *window* is not labeled to identify a monitored variable.

alert: see *process condition* and *sequence state*.

analog input point: an *alarm point* for use with an analog monitored variable signal, usually current or voltage. The logic circuit initiates an *alarm* when the analog signal is above or below a set point.

annunciator: a device or group of devices that call attention to changes in *process conditions* that have occurred. An annunciator usually calls attention to abnormal *process conditions*, but may be used also to show normal process status. Usually included are sequence logic circuits, labeled *visual displays, audible devices,* and manually operated *pushbuttons*.

audible device: a device that calls attention by sound to the occurrence of abnormal *process conditions*. An audible device may also call attention to return to normal conditions.

audible device follower: see *auxiliary output*.

automatic reset: see *reset*.

auxiliary contact: see *auxiliary output*.

auxiliary output (auxiliary contact): An output signal operated by a single *alarm point* or group of points for use with a remote device. Types of auxiliary output include:

field contact follower: an auxiliary output that operates while the *field contact* indicates an abnormal *process condition*.

lamp follower: an auxiliary output that operates while the *visual display* lamps indicate an *alarm*, silenced, or acknowledged state.

audible device follower (horn relay contact): an auxiliary output that operates while the common *alarm audible device* operates.

reflash: an auxiliary output that operates when any one of a group of *alarm points* indicates an abnormal *process condition*. The output usually returns to normal briefly when each *alarm point* changes to an abnormal *process condition* and returns to normal when all *alarm points* in the group indicate normal *process conditions*.

blank alarm point: see *alarm point*.

field contact (trouble or signal contact): the electrical contact of the device sensing the *process condition.* The contact is either open or closed. Annunciator field contacts are identified in relation to *process conditions* and annunciator operation, not the disconnected position of the devices. Types of field contact include:

normally open (NO): a field contact that is open for a normal *process condition* and closed when the *process condition* is abnormal.

normally closed (NC): A field contact that is closed for a normal *process condition* and open when the *process condition* is abnormal.

field contact follower: see *auxiliary output*.

field contact voltage (trouble or signal contact voltage): the voltage applied to *field contacts*.

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first alert: see *first out*.

first out (first alert): a *sequence* feature that indicates which of a group of *alarm points* operated first.

first out reset: see *reset*.

flasher: a device that causes *visual displays* to turn on and off repeatedly. Types of flashing include fast flashing, flashing, slow flashing, and intermittent flashing.

functional test: see *test*.

future alarm point: see *alarm point*.

horn relay contact: see *auxiliary output*.

integral logic annunciator: an *annunciator* that includes *visual displays* and sequence logic circuits in one assembly.

lamp cabinet: a cabinet containing *visual displays* only.

lamp follower: see *auxiliary output*.

lamp test: see *test*.

lock-in: a *sequence* feature that retains the alarm state until acknowledged when the abnormal *process condition* is *momentary*.

logic cabinet: a cabinet containing logic circuits and no *visual displays*.

maintained alarm: see *alarm*.

manual reset: see *reset*.

momentary alarm: see *alarm*.

multiple input: see *reflash*.

nameplate: see *window*.

normally closed (NC): see *field contact*.

normally open (NO): see *field contact.*

operational test: see *test*.

point module: see *alarm module*.

process condition: the condition of the monitored variable. The process condition is either normal or abnormal (*alarm*, alert, or off-normal).

pushbutton: A momentary manual switch that causes a change from one *sequence state* to another. Pushbutton actions include *silence, acknowledge, reset, first out reset*, and *test*.

reflash (multiple input): 1. an auxiliary logic circuit that allows two or more abnormal *process conditions* to initiate or reinitiate the *alarm state* of one *alarm point* at any time. The *alarm point* cannot return to normal until all related *process conditions* return to normal. 2. one type of *auxiliary output*.

remote logic annunciator: an *annunciator* that locates *visual displays* and sequence logic circuits in separate assemblies.

reset: the *sequence action* that returns the *sequence* to the normal state. Types of reset include:

> **automatic**: reset occurs after *acknowledge* when the *process condition* returns to normal.
>
> **manual**: reset occurs after *acknowledge* when the *process condition* has returned to normal and the reset *pushbutton* is operated.

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first out: reset of the *first out* indication occurs when the *acknowledge* or *first out* reset *pushbutton* is operated, whether the *process condition* has returned to normal or not, depending on the sequence.

response time: the time period between the *process condition* becoming abnormal and initiation of the *alarm* state. The minimum *momentary alarm* duration required for *annunciator* operation.

return alert: see *ringback*.

ringback (return alert): a *sequence* feature that provides a distinct visual or audible indication or both when the *process condition* returns to normal.

sequence: the chronological series of actions and states of an *annunciator* after an abnormal *process condition* or manual *test* initiation occurs.

sequence action: a signal that causes the *sequence* to change from one *sequence state* to another. Sequence actions include *process condition* changes and manual operation of *pushbuttons*.

sequence diagram: a graphic presentation that describes *sequence actions* and *sequence states*.

sequence module: see *alarm module*.

sequence state: the condition of the *visual display* and *audible device* provided by an *annunciator* to indicate the *process condition* or *pushbutton* actions or both. Sequence states include normal, *alarm* (alert), silenced, acknowledged, and *ringback*.

sequence table: a presentation that describes *sequence actions* and *sequence states* by lines of statements arranged in columns.

signal contact: see *field contact*.

signal contact voltage: see *field contact voltage.*

silence: the *sequence action* that stops the sound of the *audible device*.

spare alarm point: see *alarm point*.

test: an *annunciator sequence* initiated by operation of the test *pushbutton* to reveal lamp or circuit failure. Types of test include:

operational (functional): test of the *sequence, visual display* lamps, *audible devices*, and *pushbuttons*.

lamp: test of the *visual display* lamps.

trouble contact: see *field contact*.

trouble contact voltage: see *field contact voltage*.

visual display: that part of an *annunciator* or *lamp cabinet* that indicates the *sequence state*. Usually consists of an enclosure containing lamps behind a translucent *window.* The lamps can be off, flashing, or on.

window (nameplate): a component of a *visual display* made from a translucent material that is illuminated from the rear and labeled to identify the monitored variable.

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4 Sequences

4.1 Operation

Annunciators usually call attention to abnormal process conditions by the use of individual illuminated visual displays and audible devices. Annunciators may also be used to show normal process status. Changes from one annunciator sequence state to another are caused by changes in process conditions and also by manual operation of pushbuttons. The new sequence state may be dependent on the process condition that exists at the time pushbuttons are operated. Process condition changes are usually sensed by field contacts.

The visual displays usually flash to indicate abnormal process conditions and change to on when alarms are acknowledged. Additional types of flashing can indicate that process conditions have returned to normal or which of a group of alarm points operated first. All of the alarm points of an annunciator usually use the same sequence; however, different sequences can be used for individual alarm points or groups of points in one annunciator.

In this Standard, sequences making use of more than one indication device as a part of each visual display to indicate the sequence state are considered to be special because of their many variations and relatively infrequent use. Examples include the use of lamps of different colors to indicate different sequence states or which is the first of a group of alarms.

4.2 Presentation

Annunciator sequence tables describe the operation of annunciators, but often do not clearly indicate all aspects of the sequences. Examples include failure to indicate the sequence actions and states when process conditions return to abnormal again before the annunciator is reset and also when pushbuttons are operated out of the normal sequence. A sequence diagram format is used in this Standard to allow annunciator sequences to be defined completely and analyzed logically. See Figures 2 to 8.

Sequence diagrams include a block for each annunciator sequence state. The process condition, the sequence state, and the visual display and audible device conditions when in that state are indicated in each block. The blocks are arranged to describe the annunciator sequence from the normal state, through the other sequence states, and back to the normal state again. Arrows between the blocks indicate all possible sequence actions that can cause a change from one sequence state to another. Sequence actions include process condition changes and manual operation of pushbuttons.

Sequence tables are also used in this Standard since it is not always convenient to use sequence diagrams. These sequence tables are patterned after the sequence diagrams to describe all aspects of the annunciator operation. See Figures 2 to 8.

The sequence tables include a line for the initial normal state and also a line for each possible sequence action that can cause a change from one sequence state to another. The reference line numbers in the tables are identified by suffixes A and B when the new sequence state depends on the process condition that exists at the time pushbuttons are operated. References to other lines in the table are used to avoid indicating each sequence state and the related visual display and audible device conditions more than once.

When annunciators require auxiliary outputs, the output operation should be added to sequence diagrams and sequence tables or the operation should be defined by notes. The operation of auxiliary outputs during annunciator test should be defined also.

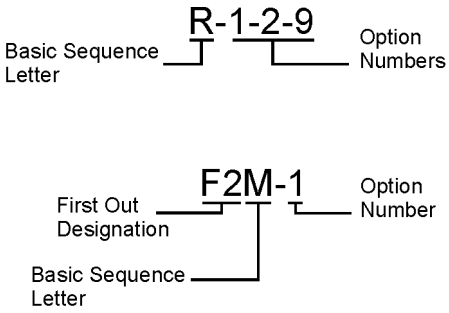
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Since most annunciators include a test pushbutton and operational test of the sequence, visual display lamps, audible devices, and pushbuttons, the sequences in this Standard include operational test as a standard feature.

4.3 Designation method

Annunciators call attention to changes in process conditions by different visual display and audible device arrangements and by a wide variety of operating sequences. The choice depends on the requirements or preferences of the users and also on the standard or special annunciator designs that are available.

This Standard provides a sequence designation method using letters for basic sequences in common use, numbers for common sequence options, and first out designations. Combinations of letters and numbers can define many different sequence variations. Sequence designation examples and a summary of the basic sequence letters, option numbers, and first out designations are provided in Figure 1. This Standard does not designate any particular sequences as being standard.



Basic Sequence Letter	Key Words
A	Automatic Reset
M	Manual Reset
R	Ringback

(See 4.4 for descriptions.)

First Out Designation	Key Words
F1	No Subsequent Alarm State
F2	No Subsequent Alarm Flashing
F3	First Out Flashing and Reset Pushbutton

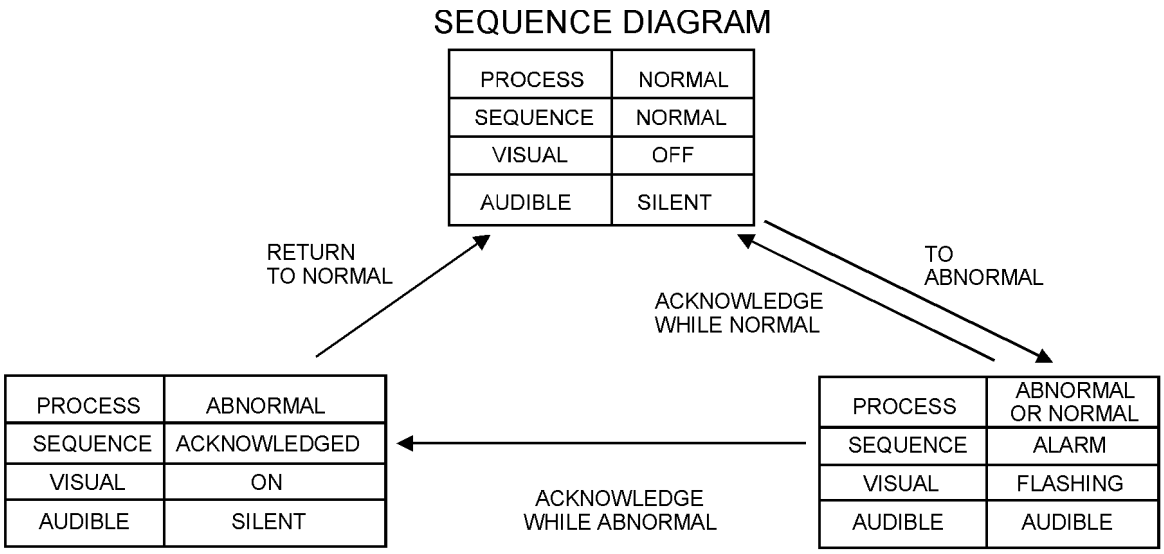
(See 4.6 for descriptions.)

Option Number	Key Words
1	Silence Pushbutton
2	Silence Interlock
3	First Out Reset Interock
4	No Lock-In
5	No Flashing
6	No Audible
7	Automatic Alarm Silence
8	Common Ringback Audible
9	Automatic Ringback Silence
10	No Ringback Audible
11	Common Ringback Visual
12	Automatic Momentary Ringback
13	Dim Lamp Monitor
14	Lamp Test

(See 4.5 for descriptions.)

Figure 1 — Annunciator sequence designations

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SEQUENCE TABLE

LINE	PROCESS CONDITION	PUSHBUTTON OPERATION	SEQUENCE STATE	VISUAL DISPLAY	ALARM AUDIBLE DEVICE	REMARKS
1	NORMAL	-	NORMAL	OFF	SILENT	
2	ABNORMAL	-	ALARM	FLASHING	AUDIBLE	LOCK-IN
3A	ABNORMAL	ACKNOWLEDGE	ACKNOWLEDGED	ON	SILENT	MAINTAINED ALARM
3B	NORMAL	ACKNOWLEDGE	TO LINE 4			MOMENTARY ALARM
4	NORMAL	-	NORMAL	OFF	SILENT	AUTOMATIC RESET

SEQUENCE FEATURES

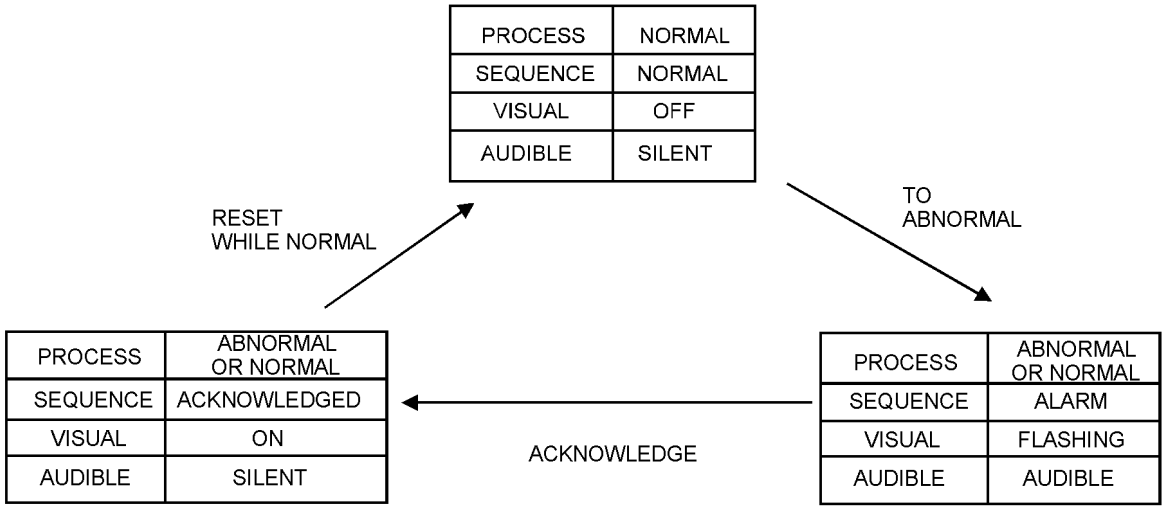
1 - ACKNOWLEDGE AND TEST PUSHBUTTONS.
2 - ALARM AUDIBLE DEVICE.
3 - LOCK-IN OF MOMENTARY ALARMS UNTIL ACKNOWLEDGED.
4 - THE AUDIBLE DEVICE IS SILENCED AND FLASHING STOPS WHEN ACKNOWLEDGED.
5 - AUTOMATIC RESET OF ACKNOWLEDGED ALARM INDICATIONS WHEN PROCESS CONDITIONS RETURN TO NORMAL.
6 - OPERATIONAL TEST.

Figure 2 — Sequence A, automatic reset

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SEQUENCE DIAGRAM



SEQUENCE TABLE

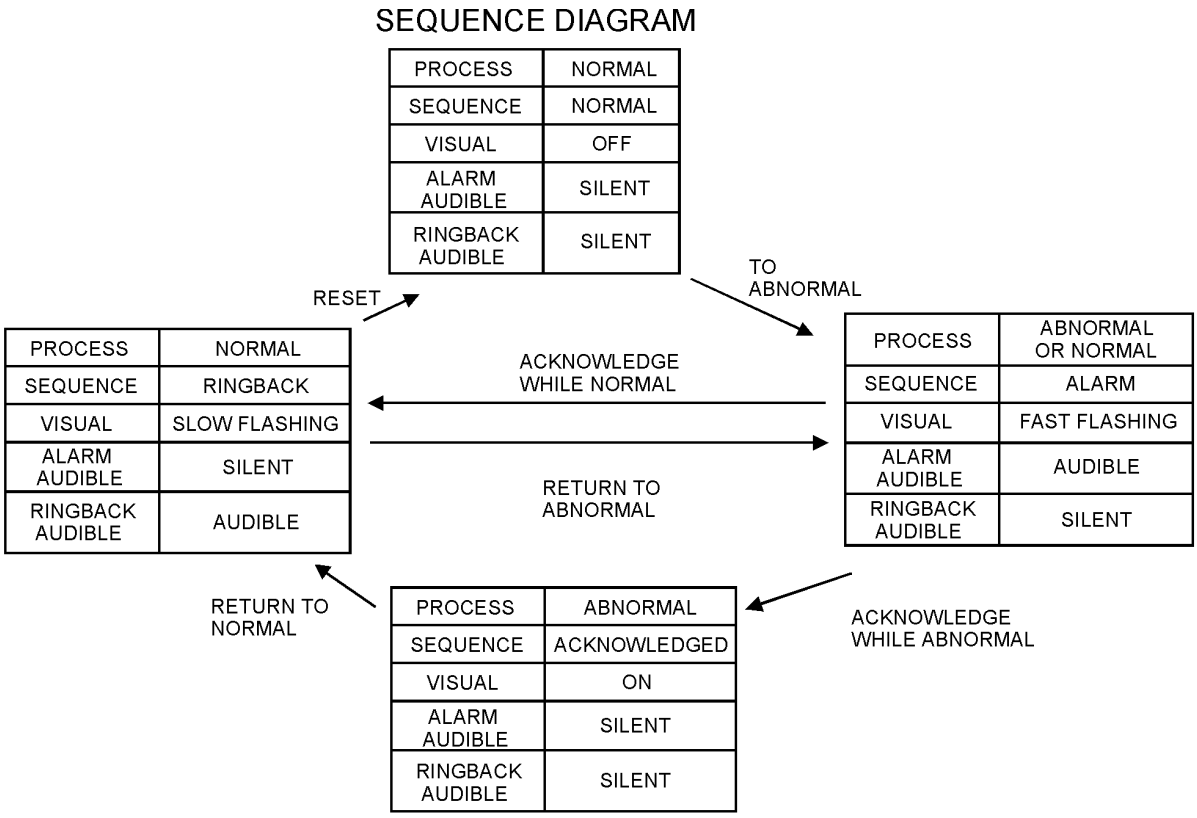
LINE	PROCESS CONDITION	PUSHBUTTON OPERATION	SEQUENCE STATE	VISUAL DISPLAY	ALARM AUDIBLE DEVICE	REMARKS
1	NORMAL	—	NORMAL	OFF	SILENT	
2	ABNORMAL	—	ALARM	FLASHING	AUDIBLE	LOCK-IN
3	ABNORMAL OR NORMAL	ACKNOWLEDGE	ACKNOWLEDGED	ON	SILENT	MANUAL RESET REQUIRED
4A	ABNORMAL	RESET	TO LINE 3			
4B	NORMAL	RESET	NORMAL	OFF	SILENT	MANUAL RESET

SEQUENCE FEATURES

1 - ACKNOWLEDGE, RESET, AND TEST PUSHBUTTONS.
2 - ALARM AUDIBLE DEVICE.
3 - LOCK-IN OF MOMENTARY ALARMS UNTIL ACKNOWLEDGED.
4 - THE AUDIBLE DEVICE IS SILENCED AND FLASHING STOPS WHEN ACKNOWLEDGED.
5 - MANUAL RESET OF ACKNOWLEDGED ALARM INDICATIONS AFTER PROCESS CONDITIONS RETURN TO NORMAL.
6 - OPERATIONAL TEST.

Figure 3 — Sequence M, manual reset

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SEQUENCE TABLE

LINE	PROCESS CONDITION	PUSHBUTTON OPERATION	SEQUENCE STATE	VISUAL DISPLAY	ALARM AUDIBLE DEVICE	RINGBACK AUDIBLE DEVICE	REMARKS
1	NORMAL	—	NORMAL	OFF	SILENT	SILENT	
2	ABNORMAL	—	ALARM	FAST FLASHING	AUDIBLE	SILENT	LOCK-IN
3A	ABNORMAL	ACKNOWLEDGE	ACKNOWLEDGED	ON	SILENT	SILENT	MAINTAINED ALARM
3B	NORMAL	ACKNOWLEDGE	TO LINE 4				MOMENTARY ALARM
4	NORMAL	—	RINGBACK	SLOW FLASHING	SILENT	AUDIBLE	MANUAL RESET REQUIRED
5	ABNORMAL	—	TO LINE 2				RETURN TO ABNORMAL
6	NORMAL	RESET	NORMAL	OFF	SILENT	SILENT	MANUAL RESET

SEQUENCE FEATURES

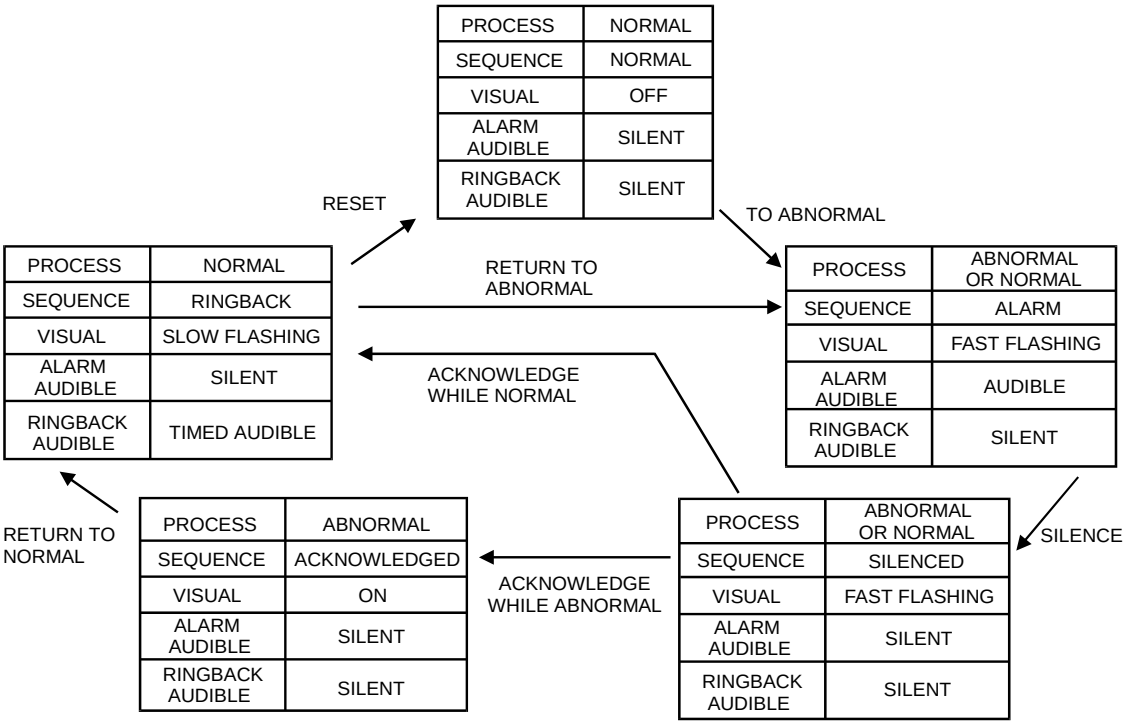
1 - ACKNOWLEDGE, RESET, AND TEST PUSHBUTTONS.
2 - ALARM AND RINGBACK AUDIBLE DEVICES.
3 - LOCK-IN OF MOMENTARY ALARMS UNTIL ACKNOWLEDGED.
4 - THE AUDIBLE DEVICE IS SILENCED AND FAST FLASHING STOPS WHEN ACKNOWLEDGED.
5 - RINGBACK VISUAL AND AUDIBLE INDICATIONS WHEN PROCESS CONDITIONS RETURN TO NORMAL.
6 - MANUAL RESET OF RINGBACK INDICATIONS.
7 - OPERATIONAL TEST.

Figure 4 — Sequence R, ringback

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SEQUENCE DIAGRAM



SEQUENCE TABLE

LINE	PROCESS CONDITION	PUSHBUTTON OPERATION	SEQUENCE STATE	VISUAL DISPLAY	ALARM AUDIBLE DEVICE	RINGBACK AUDIBLE DEVICE	REMARKS
1	NORMAL	—	NORMAL	OFF	SILENT	SILENT	
2	ABNORMAL	—	ALARM	FAST FLASHING	AUDIBLE	SILENT	LOCK-IN
3	ABNORMAL OR NORMAL	SILENCE	SILENCED	FAST FLASHING	SILENT	SILENT	LOCK-IN
4A	ABNORMAL	ACKNOWLEDGE	ACKNOWLEDGED	ON	SILENT	SILENT	MAINTAINED ALARM
4B	NORMAL	ACKNOWLEDGE	TO LINE 5				MOMENTARY ALARM
5	NORMAL	—	RINGBACK	SLOW FLASHING	SILENT	TIMED AUDIBLE	MANUAL RESET REQUIRED
6	ABNORMAL	—	TO LINE 2				RETURN TO ABNORMAL
7	NORMAL	RESET	NORMAL	OFF	SILENT	SILENT	MANUAL RESET

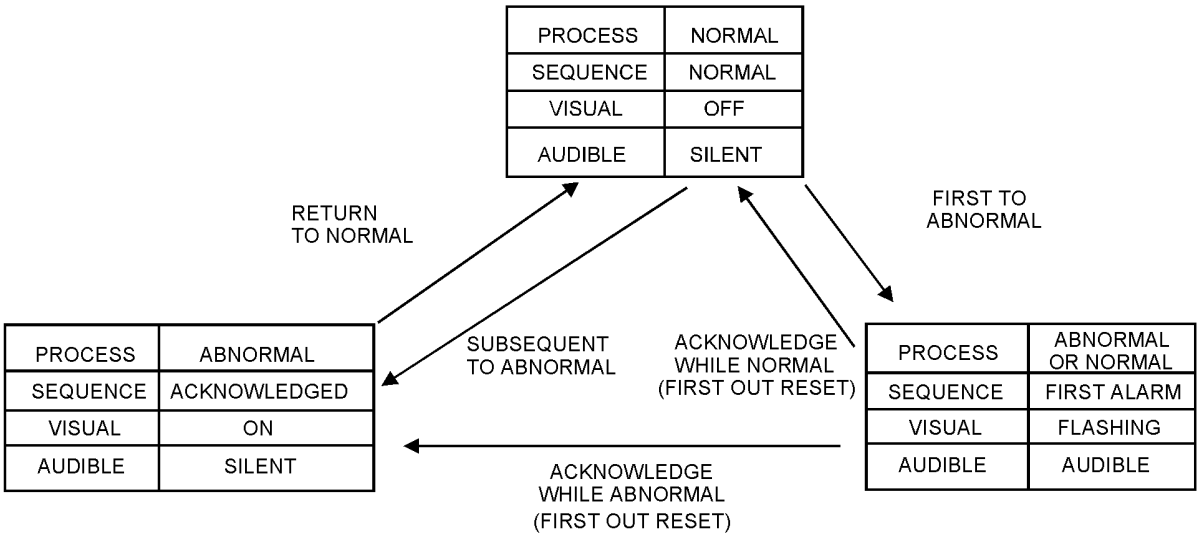
SEQUENCE FEATURES

1 - SILENCE, ACKNOWLEDGE, RESET, AND TEST PUSHBUTTONS.
2 - ALARM AND RINGBACK AUDIBLE DEVICES.
3 - LOCK-IN OF MOMENTARY ALARMS UNTIL ACKNOWLEDGED.
4 - OPTION 1 - SILENCE PUSHBUTTON TO SILENCE THE ALARM AUDIBLE DEVICE WHILE RETAINING FAST FLASHING INDICATIONS.
5 - OPTION 2 - SILENCE INTERLOCK TO REQUIRE OPERATION OF THE SILENCE PUSHBUTTON BEFORE THE ACKNOWLEDGE PUSHBUTTON.
6 - RINGBACK VISUAL AND AUDIBLE INDICATIONS WHEN PROCESS CONDITIONS RETURN TO NORMAL.
7 - OPTION 9 - AUTOMATIC RINGBACK SILENCE TO SILENCE THE RINGBACK AUDIBLE DEVICE AFTER A SET TIME.
8 - MANUAL RESET OF RINGBACK INDICATIONS.
9 - OPERATIONAL TEST.

Figure 5 — Sequence R-1-2-9, ringback with options

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SEQUENCE DIAGRAM



SEQUENCE TABLE

LINE	PROCESS CONDITION		PUSHBUTTON OPERATION	SEQUENCE STATE	VISUAL DISPLAY	ALARM AUDIBLE DEVICE	REMARKS
1	NORMAL		—	NORMAL	OFF	SILENT	
2	FIRST	ABNORMAL	—	FIRST ALARM	FLASHING	AUDIBLE	LOCK-IN
3	SUB.	ABNORMAL	—	ACKNOWLEDGED	ON	SILENT	NO LOCK-IN
4A	FIRST	ABNORMAL	ACKNOWLEDGE	TO LINE 3			MAINTAINED ALARM FIRST OUT RESET
4B	FIRST	NORMAL	ACKNOWLEDGE	TO LINE 5			MOMENTARY ALARM FIRST OUT RESET
5	NORMAL		—	NORMAL	OFF	SILENT	AUTOMATIC RESET

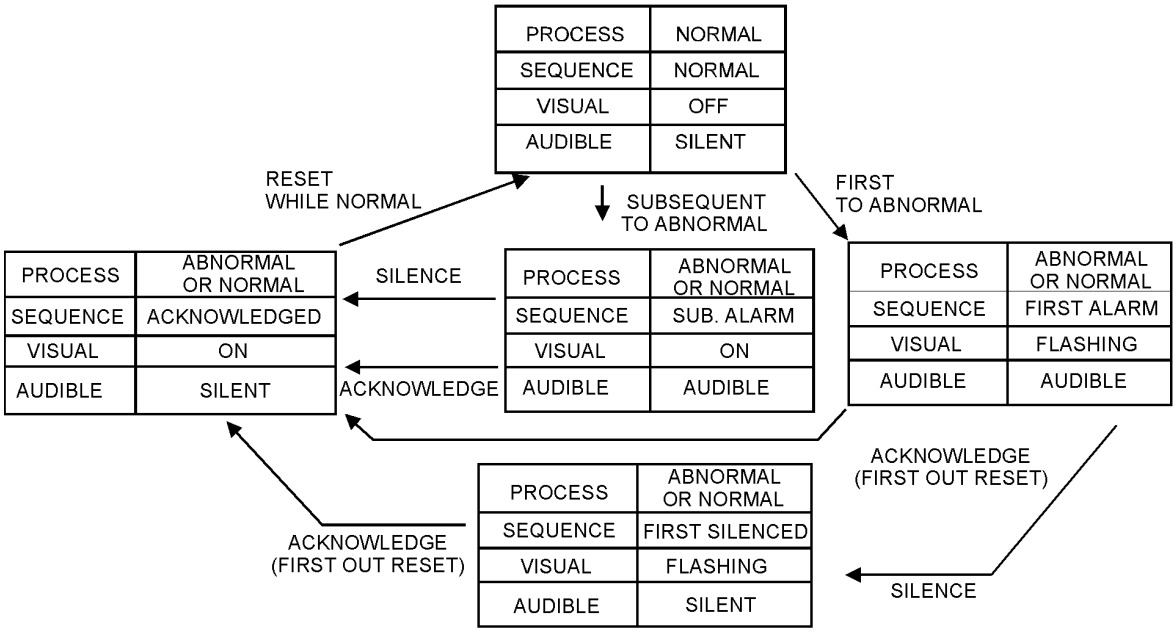
SEQUENCE FEATURES

1 - ACKNOWLEDGE AND TEST PUSHBUTTONS.
2 - ALARM AUDIBLE DEVICE.
3 - LOCK-IN OF MOMENTARY FIRST ALARM UNTIL ACKNOWLEDGED. NO LOCK-IN OF MOMENTARY SUBSEQUENT ALARMS.
4 - FLASHING AND AUDIBLE INDICATIONS FOR FIRST ALARM ONLY. NEW SUBSEQUENT ALARMS GO TO THE ACKNOWLEDGE STATE.
5 - FIRST OUT INDICATION IS RESET AND THE AUDIBLE IS SILENCED WHEN ACKNOWLEDGED.
6 - AUTOMATIC RESET OF ACKNOWLEDGED ALARM INDICATIONS WHEN PROCESS CONDITIONS RETURN TO NORMAL.
7 - OPERATIONAL TEST.

Figure 6 — Sequence F1A, automatic reset first out with no subsequent alarm state

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SEQUENCE DIAGRAM



SEQUENCE TABLE

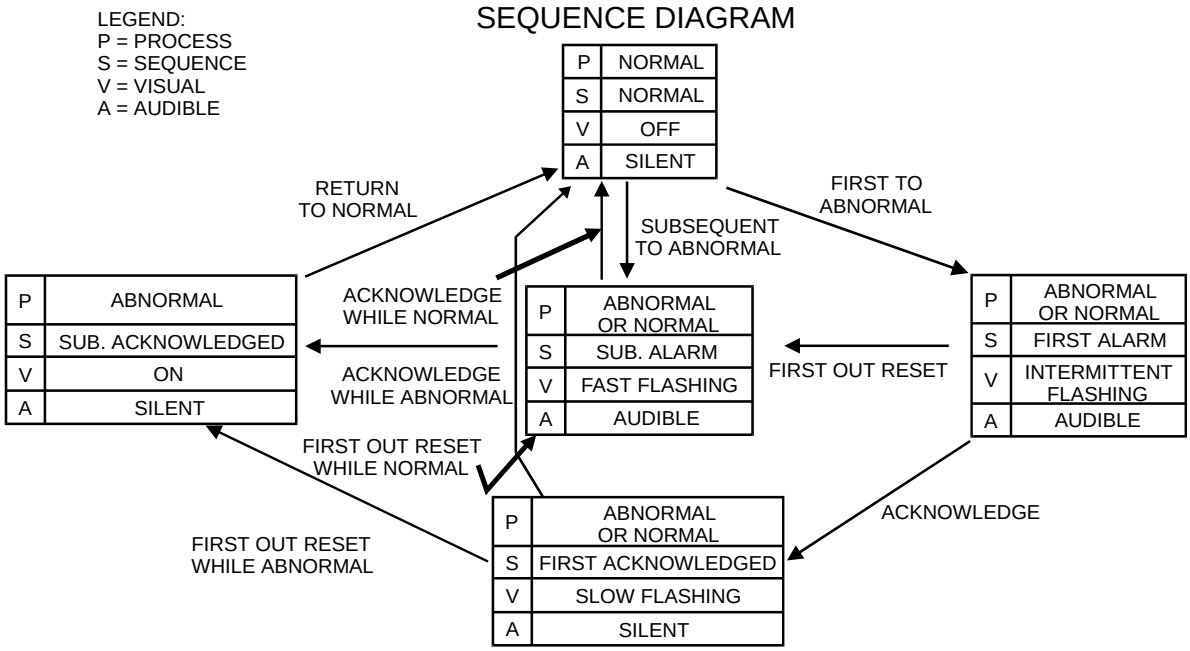
LINE	PROCESS CONDITION		PUSHBUTTON OPERATION	SEQUENCE STATE	VISUAL DISPLAY	ALARM AUDIBLE DEVICE	REMARKS
1	NORMAL		—	NORMAL	OFF	SILENT	
2	FIRST	ABNORMAL	—	FIRST ALARM	FLASHING	AUDIBLE	LOCK-IN
3	SUB.	ABNORMAL	—	SUB. ALARM	ON	AUDIBLE	LOCK-IN
4	FIRST	ABNORMAL OR NORMAL	ACKNOWLEDGE BEFORE SILENCE	TO LINE 7			FIRST OUT RESET
5	SUB.	ABNORMAL OR NORMAL					
6	FIRST	ABNORMAL OR NORMAL	SILENCE	FIRST SILENCED	FLASHING	SILENT	
7	SUB.	ABNORMAL OR NORMAL	SILENCE	ACKNOWLEDGE	ON	SILENT	MANUAL RESET REQUIRED
8	FIRST	ABNORMAL OR NORMAL	ACKNOWLEDGE AFTER SILENCE	TO LINE 7			FIRST OUT RESET
9	NORMAL		RESET	NORMAL	OFF	SILENT	MANUAL RESET

SEQUENCE FEATURES

1 - SILENCE, ACKNOWLEDGE, RESET, AND TEST PUSHBUTTONS.
2 - ALARM AUDIBLE DEVICE.
3 - LOCK-IN OF MOMENTARY ALARMS UNTIL ACKNOWLEDGED.
4 - OPTION 1 - SILENCE PUSHBUTTON TO SILENCE THE ALARM AUDIBLE DEVICE WHILE RETAINING FIRST OUT FLASHING INDICATION.
5 - FLASHING INDICATION FOR FIRST ALARM ONLY. NEW SUBSEQUENT ALARMS HAVE SAME VISUAL INDICATION AS ACKNOWLEDGED ALARMS.
6 - FIRST OUT INDICATION IS RESET WHEN ACKNOWLEDGED.
7 - MANUAL RESET OF ACKNOWLEDGED ALARM INDICATIONS AFTER PROCESS CONDITIONS RETURN TO NORMAL.
8 - OPERATIONAL TEST.

Figure 7 — Sequence F2M-1, manual reset first out with no subsequent alarm flashing and silence pushbutton

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SEQUENCE TABLE

LINE	PROCESS CONDITIONS		PUSHBUTTON OPERATION	SEQUENCE STATE	VISUAL DISPLAY	ALARM AUDIBLE DEVICE	REMARKS
1	NORMAL		—	NORMAL	OFF	SILENT	
2	FIRST	ABNORMAL	—	FIRST ALARM	INTERMITTENT FLASHING	AUDIBLE	LOCK-IN
3	SUB.	ABNORMAL	—	SUB. ALARM	FAST FLASHING	AUDIBLE	LOCK-IN
4	FIRST	ABNORMAL OR NORMAL	FIRST OUT RESET BEFORE ACKNOWLEDGE	TO LINE 3			FIRST OUT RESET
5	FIRST	ABNORMAL OR NORMAL	ACKNOWLEDGE	FIRST ACKNOWLEDGED	SLOW FLASHING	SILENT	FIRST OUT RESET REQUIRED
6A	SUB.	ABNORMAL		SUB. ACKNOWLEDGED	ON	SILENT	MAINTAINED ALARM
6B	SUB.	NORMAL		TO LINE 8			MOMENTARY ALARM
7A	FIRST	ABNORMAL	FIRST OUT RESET AFTER ACKNOWLEDGE	TO LINE 6A			FIRST OUT RESET
7B	FIRST	NORMAL		TO LINE 8			FIRST OUT RESET
8	NORMAL		—	NORMAL	OFF	SILENT	AUTOMATIC RESET

SEQUENCE FEATURES

1 - ACKNOWLEDGE, FIRST OUT RESET, AND TEST PUSHBUTTONS.
2 - ALARM AUDIBLE DEVICE.
3 - LOCK-IN OF MOMENTARY ALARMS UNTIL ACKNOWLEDGED.
4 - FIRST OUT FLASHING DIFFERENT FROM SUBSEQUENT FLASHING.
5 - FIRST OUT RESET PUSHBUTTON TO CHANGE THE FIRST OUT VISUAL INDICATION TO BE THE SAME AS SUBSEQUENT VISUAL INDICATIONS.
6 - AUTOMATIC RESET OF ACKNOWLEDGED ALARM INDICATIONS WHEN PROCESS CONDITIONS RETURN TO NORMAL.
7 - OPERATIONAL TEST.

Figure 8 — Sequence F3A, automatic reset first out with first out flashing and reset pushbutton

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4.4 Basic sequence letter designations

Three basic types of annunciator sequence are in common use. The operation of each is different after process conditions return to normal. This Standard makes use of the following basic sequence letters to designate the three basic sequence types:

Basic Sequence Letter	Key Words	Description
A	Automatic Reset	The sequence returns to the normal state automatically after acknowledge when the process condition returns to normal.
M	Manual Reset	The sequence returns to the normal state after acknowledge when the process condition has returned to normal and the reset pushbutton is operated.
R	Ringback	The sequence provides distinct visual and audible indications when the process condition returns to normal. The sequence returns to normal after acknowledge when the process condition has returned to normal and the reset pushbutton is operated.

Sequence diagrams and sequence tables for these basic sequences are shown in Figures 2, 3, and 4.

The types of flashing shown in this Standard such as fast flashing and slow flashing are examples based on frequent use. Alternate types of flashing may be used without requiring a change in the sequence designation.

Since annunciator sequences usually include lock-in of momentary alarms, sequences in this Standard include lock-in as a standard feature. A sequence option number is provided to permit deleting the lock-in feature. Some alarm modules have provisions for deleting the lock-in feature on individual alarm points.

Variations in the basic sequences can be defined using basic sequence letter designations combined with option numbers—see 4.5, Option Number Designations.

First out sequences require a first out designation in addition to the basic sequence letter designation—see 4.6, First Out Designations.

4.5 Option number designations

Option numbers can be used with the basic sequence letter designations to define many different sequence variations. This Standard makes use of the following option numbers to designate many of the common sequence variations. Other sequence variations are considered to be special and should be defined by sequence diagrams, sequence tables, or notes.

An example of a sequence designation with option numbers is shown in Figure 1. Figures 5 and 7 illustrate the use of option numbers.

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Option Number	Key Words	Description
1	Silence Pushbutton	A separate pushbutton is added to allow silencing the alarm audible device without affecting the visual displays.
2	Silence Interlock	An interlock is added to require operation of the silence pushbutton before alarms can be acknowledged.
3	First Out Reset Interlock	An interlock is added to require operation of the acknowledge pushbutton before first out alarms can be reset by the first out reset pushbutton.
4	No Lock-In	The lock-in feature is deleted. Momentary alarms return to the normal sequence state without operation of the acknowledge pushbutton.
5	No Flashing	The visual display flashing feature is deleted. New alarms have the same visual display indication as acknowledged alarms.
6	No Audible	The audible device is deleted.
7	Automatic Alarm Silence	A time delay device is added to silence the alarm audible device after a set time without affecting the visual displays.
8	Common Ringback Audible	A common audible device is used to call attention to both the alarm and ringback sequence states.
9	Automatic Ringback Silence	A time delay device is added to silence the ringback audible device after a set time without affecting the visual displays.
10	No Ringback Audible	The ringback audible device is deleted.
11	Common Ringback Visual	A common type of flashing is used to indicate both the alarm and ringback sequence states.
12	Automatic Momentary Ringback	Ringback sequence momentary alarms go to the ringback sequence state without operation of the acknowledge pushbutton.
13	Dim Lamp Monitor	The visual display indication is dim in the normal sequence state to reveal lamp failure.
14	Lamp Test	Operation of the test pushbutton tests the visual displays only.

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4.6 First out designations

First out annunciators are used to indicate which one of a group of alarm points operated first. To accomplish this, the visual display indication for the alarm point that operates first must be different from the visual display indication for subsequent alarm points in that group. Only one first out alarm indication can exist in any one first out group.

Three methods for differentiating between first and subsequent alarms are in common use. Two make use of the usual sequence features for the first alarm and delete features for subsequent alarms. The third provides additional features to indicate first alarms. This Standard makes use of the following first out designations to designate the three methods.

First Out Designation	Key Words	Description
F1	No Subsequent Alarm State	Subsequent alarms appear in the acknowledged state. Subsequent visual displays do not flash. The audible device does not operate when subsequent alarms occur, unless still operating from the first alarm. The first out indication is reset by the acknowledge pushbutton.
F2	No Subsequent Alarm Flashing	Subsequent visual quent displays do not flash. The audible device does not operate when subsequent alarms occur. The first out indication is reset by the acknowledge pushbutton.
F3	First Out Flashing and Reset Pushbutton	Additional types of flashing are added to identify new and acknowledged first alarms. A first out reset pushbutton is added to reset the first out indication, whether the process condition has returned to normal or not.

First out sequences can be automatic reset or manual reset or can provide ringback indication when alarms return to normal. First out sequences are designated by a combination of the first out designation, the basic sequence letter designation, and option numbers. An example of a first out sequence designation is in Figure 1.

First out sequence diagrams consist of an outer loop of actions and states associated with the first alarm and an inner loop associated with subsequent alarms. The two loops have a common normal state.

Not all of the possible first out sequences are readily available. In some cases, a particular first out sequence may be a standard design for only one manufacturer. Sequence designations for a range of first out sequences are listed below. Some of these use a silence pushbutton, option number 1, to silence the audible device while retaining the visual display indications in the alarm state—see 4.5, Option Number Designations. The sequences commonly available at the time of publication are indicated. Sequence diagrams and sequence tables for three of the common first out sequences are shown in Figures 6, 7, and 8.

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Key Words	Automatic Reset	Manual Reset	Ringback
No Subsequent Alarm State	F1A (Common) (Figure 6)	F1M (Common)	F1R
No Subsequent Alarm State and Silence Pushbutton	F1A-1 (Common)	F1M-1 (Common)	F1R-1
No Subsequent Alarm Flashing	F2A (Common)	F2M (Common)	F2R
No Subsequent Alarm Flashing and Silence Pushbutton	F2A-1 (Common)	F2M-1 (Common) (Figure 7)	F2R-1
First Out Flashing and Reset Pushbutton	F3A (Common) (Figure 8)	F3M	F3R

5 Specifications

Annunciator specifications provide manufacturers with the information necessary to prepare proposals and to design and produce the required annunciator equipment. Many details of annunciator design are relatively standard and may not need to be specified. Some annunciators require only standard specification forms* or brief specifications. Complex annunciator systems generally require more elaborate specifications to define the system requirements.

When preparing annunciator specifications, careful thought must be given when specifying features that are not readily available. The advantages of such features should be weighed against the disadvantages of special design.

The following types of information should be included in annunciator specifications, but other features should also be specified as required.

5.1 All annunciators

1) Logic circuit location: integral logic, remote logic cabinet

2) Sequence: ISA designation, other identification, sequence diagram, sequence table, notes, arrangement when more than one sequence is used

3) Number of alarm points: total, active, spare, future

4) Power source: nominal voltage, frequency

*Refer to ISA Standard ISA-S20, “Specification Forms for Process Measurement and Control Instruments, Primary Elements, and Control Valves.”

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5) Nominal window size: dimension high, dimension wide

6) Visual display arrangement: rows high, columns wide

7) Cabinet mounting: flush, surface, rack

8) Cabinet type: NEMA enclosure type or electrical classification

9) Window engraving: legend list, lettering size, maximum number of lines, maximum characters per line (May also be provided later.)

10) Logic circuit type: solid state, electromechanical relay

11) Field contact operation: normally open (close to alarm), normally closed (open to alarm), mixed

12) Field contact voltage: nominal voltage, frequency

13) Information required with proposal: descriptions, drawings, price, delivery

14) Documentation required: after award, before delivery, with delivery

5.2 Remote logic annunciators

1) Logic cabinet mounting: surface, chassis, rack, freestanding

2) Logic cabinet type: NEMA enclosure type or electrical classification

3) Logic cabinet requirements: arrangement, cable entrance, color

4) Prefabricated cables: supplier, length, conductor size, insulation type, jacket type

5) Lamp cabinet connections: plug connectors, screw terminals

6) Logic cabinet connections to lamp cabinet: plug connectors, screw terminals

5.3 Complex annunciator systems

1) System arrangement drawing: cabinets, prefabricated cables, pushbuttons, audible devices, power sources

2) System operation description

5.4 Annunciator accessories and special features

1) Pushbuttons: location, supplier, number, type

2) Audible devices: location, supplier, power, type

3) Special cabinet finish: annunciator or lamp cabinet color, logic cabinet color, materials to be used, application methods

4) Special visual display: individual windows, graphic displays

5) Special window or window bezel colors: colors, windows

6) Reflash points: number of field contacts, alarm points

7) Special field contact time delay: time delay, alarm points

8) Analog input points: analog signals, alarm points

9) Auxiliary outputs: function, operation on test, grouping, type, electrical rating

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10) Solid state field contacts or solid state auxiliary outputs: type, electrical rating, common potential

11) Special power source requirements: voltage variation range, frequency variation range

12) Backup power system: power sources, requirements

13) Isolation from power sources: field contacts, window lamps, logic circuits

14) Special power source for receptacles and lights: nominal voltage, frequency

15) Power failure detector: alarm, indication

16) Ground detector: alarm, indication, isolation switches

17) Window legibility: lettering size, ambient light

18) Special service conditions: temperature, humidity, ambient light, noise, hazardous atmosphere, intrinsically safe design, nonincendive design, purged cabinet, high field contact wiring impedance

19) Special wire insulation: voltage rating, insulation type, test requirements

20) Special field contact wiring terminations: terminal type, terminal size, wiring space

21) Special grounding connections: terminal size, ground bus

22) Special cabinet locks: type, keys

23) Special factory tests: dielectric strength, functional, surge with stand capability, radio frequency interference, seismic, nuclear Class 1E

24) Special documentation: test procedures, test reports

25) Spare components: type, number

26) Logic circuit tester: portable, built-in

6 Documentation

Documentation is provided by manufacturers to describe the equipment produced to meet annunciator specifications. Before delivery of annunciators, this documentation is used to confirm that specification requirements are met and to allow design of the annunciator mounting, external wiring, and power sources. After delivery, this documentation is used by those installing, operating, and maintaining annunciators.

The following types of information should be included in the documentation provided by the manufacturer, but other documentation should be provided as required by specifications.

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6.1 All annunciators

1) Annunciator description: appearance, standard features, special features

2) Dimensioned drawings: outline, arrangement, enclosure type, panel cutout, mounting method, wiring entrance

3) Sequence description: ISA designation, sequence diagram, sequence table, notes

4) Schematic diagram drawings: logic circuit, circuit wiring, jumper and switch settings, modules, pushbuttons, audible devices, field contact voltage, lamp voltage, power supply circuit, electrical ratings

5) Wiring diagram drawings: external wiring, power source voltage and frequency

6) Power requirements: normal, maximum, allowable voltage and frequency variation ranges

7) Instruction manual: installation, operation, operating limitations, logic circuit operation, maintenance procedures, special test device operation

8) Parts list: replacement parts, recommended spare parts, prices

6.2 Remote logic annunciators

1) Logic cabinet arrangement drawings: internal devices, terminal blocks, wiring space, ground connections, receptacles, lights, fans

2) Interconnection drawings: external wiring connections between cabinets

3) Power supply drawings: power supply and distribution system

6.3 Complex annunciator systems

1) System arrangement drawings: components, interconnections

2) System operation description

6.4 Annunciator accessories and special features

Necessary information concerning accessories and special features should be provided with the annunciator documents.

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Appendix A — Annunciator application guide

This Appendix is included for informational purposes and is not a part of this Standard.

A.1 Purpose

The purpose of this Appendix is to provide information to assist in specifying annunciators that will best serve the requirements of the users while making use of standard and special features that are readily available. Catalogs and other information from annunciator manufacturers should also be used because of the wide variety of features that are available.

A.2 Introduction

Annunciators are normally used to call attention to abnormal process conditions. Annunciators may also be used to show normal process status. Annunciators usually include individual illuminated visual displays that are labeled to identify the particular monitored variable that is abnormal and audible devices. Annunciators may also call attention to the return to normal of the process conditions.

Visual displays usually flash to indicate abnormal process conditions. Manual operation of pushbuttons is usually required to silence audible devices and acknowledge new alarms. Visual displays usually change from flashing to on when alarms are acknowledged. Figure A.1 illustrates a typical annunciator sequence.

Additional types of flashing can indicate that process conditions have returned to normal or which of a group of alarm points operated first. Additional pushbuttons can be used to acknowledge alarms that return to normal, to reset first out indications, and to test annunciator lamps and circuits.

Annunciators are available in an almost infinite variety of physical arrangements, operating sequences, and special features. In some cases, the annunciation function is performed by computer systems using electronic screen displays or recording annunciators—see A.5, Special Annunciators.

This Standard is primarily for use with electrical annunciators that use illuminated visual displays and audible devices. Enclosures with lamps behind labeled translucent windows are commonly used as visual displays. Annunciators usually operate from electrical contacts that are part of the devices that sense the process conditions.

In this Standard, sequences making use of more than one indication device as a part of each visual display to indicate the sequence state are considered to be special because of their many variations and relatively infrequent use. When additional wiring and sockets are provided, colored lights such as red and green may be used along with flashing to indicate sequence states or which is the first of a group of alarms. Colored lights may also be used to uniquely identify some of the alarms in an annunciator—see A.8.2, Windows.

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A.3.1 Designation method

The sequence designation method provided in this Standard uses letters for basic sequences in common use, numbers for common options, and first out designations. Combinations of letters and numbers can define many different sequence variations. Refer to Figure 1, of this Standard for sequence designation examples and a summary of the basic sequence letters, option numbers, and first out designations.

This Standard does not designate any particular sequences as being standard. Refer to Appendix B, Sequence Designation Conversion, for the ISA sequences from ISA Recommended Practice ISA-RP18.1-1965 that are superseded and the corresponding new sequence designations.

Sequence diagrams, sequence tables, or notes should be used when sequence letter and option number designations do not adequately define required sequences.

When selecting sequence designations, careful thought must be given when including features or combinations of features that are not readily available. The advantages of such features should be weighed against the disadvantages of special design.

A.3.2 Lock-in

Lock-in is a sequence feature that retains the alarm state until acknowledged when momentary alarms occur. With lock-in, momentary alarms can be observed before the acknowledge pushbutton is operated. Without lock-in, momentary alarms may not be observed at all, even though the audible device operates briefly.

Since annunciator sequences usually include lock-in, the sequences in this Standard include lock-in as a standard feature. Option Number 4 is provided to permit deleting the lock-in feature—see 4.5, Option Number Designations. Some alarm modules include movable jumpers or switches to allow deleting the lock-in feature on individual alarm points.

When momentary alarms occur frequently enough to be a nuisance, the lock-in feature is often deleted to avoid having to acknowledge the alarms repeatedly. In such cases, a better solution may be to correct the cause of the momentary alarms or to use time delays in field contact circuits in order to alarm only abnormal process conditions that exist longer than a given time.

A.3.3 Sequence A, automatic reset

Sequence A is a basic annunciator sequence with automatic reset that automatically returns acknowledged alarms to normal when process conditions return to normal. A sequence diagram and sequence table for sequence A are shown in Figure 2, of this Standard.

In some applications, sequence A may have a disadvantage since new momentary alarms return to off and new maintained alarms change to on when and during the time the audible device is silenced by operation of the acknowledged pushbutton. New alarms may be lost or may be confused with existing acknowledged alarms. New alarms may have to be reviewed or logged while flashing, with the continual distraction of the audible device signal. If these features are not desirable, sequence A-1 with a silence pushbutton or sequence A-1-2 with a silence pushbutton and interlock should be used—see 4.5, Option Number Designations.

If an audible signal is required when process conditions return to normal, sequence R that includes ringback should be used—see A.3.5, Sequence R, Ringback.

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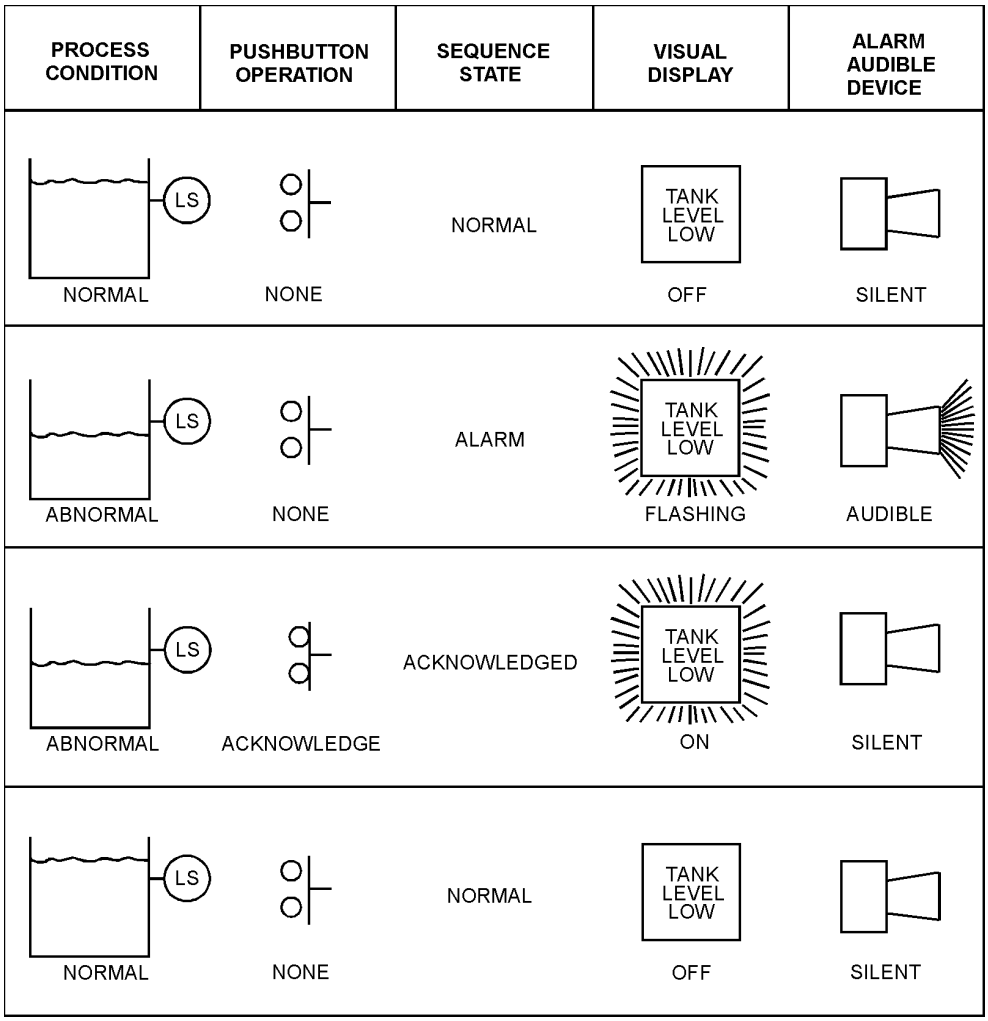


Figure A.1 — Typical annunciator sequence, sequence A, automatic reset

A.3.4 Sequence M, manual reset

Sequence M is a basic annunciator sequence with manual reset that retains acknowledged alarms until the process conditions return to normal and the manual reset pushbutton is operated. A sequence diagram and sequence table for sequence M are shown in Figure 3 of this Standard.

In some applications, sequence M may have a disadvantage since new alarms change to on when and during the time the audible device is silenced by operation of the acknowledge pushbutton. New alarms may be confused with existing acknowledged alarms. New alarms may have to be reviewed or logged while flashing, with the continual distraction of the audible device signal. If these features are not desirable, sequence M-1 with a silence pushbutton or sequence M-1-2 with silence pushbutton and interlock should be used—see 4.5, Option Number Designations.

In order to reset alarms, sequence M requires that the reset pushbutton be operated repeatedly to determine if process conditions have returned to normal. When the reset pushbutton is operated, it may be difficult to observe which of a number of acknowledged alarms have returned to normal. With sequence M, it is not evident when process conditions return to normal or return again to abnormal. If these features are not desirable, sequence R that includes ringback should be used—see A.3.5, Sequence R, Ringback.

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Sequence M usually cannot be used with reflash circuits or to operate from remote reflash auxiliary outputs since the sequence does not return to the alarm state when the field contact circuit returns to normal briefly. Refer to A.9.3, Shared Alarm Points, and A.9.5, Auxiliary Outputs. If this feature is required, sequence A or R should be used—see A.3.3, Sequence A, Automatic Reset and A.3.5, Sequence R, Ringback.

A.3.5 Sequence R, ringback

Sequence R is a basic annunciator sequence with ringback that provides distinct visual and audible indications when process conditions return to normal. The ringback indications are retained until the process conditions return to normal and the manual reset pushbutton is operated. A sequence diagram and sequence table for sequence R are shown in Figure 4 of this Standard.

In some applications, sequence R may have a disadvantage since new momentary alarms change to slow flashing and new maintained alarms change to on when and during the time the audible device is silenced by operation of the acknowledge pushbutton. New alarms may be confused with existing alarms. New alarms may have to be reviewed or logged while fast flashing, with the continual distraction of the alarm audible device signal. If these features are not desirable, sequence R-1 with a silence pushbutton or sequence R-1-2 with silence pushbutton and interlock should be used—see 4.5, Option Number Designations.

Sequence R includes different visual display indications and different audible device signals for alarm and ringback. Several variations of this arrangement can be used. Sequence R-8 uses a common audible device for both alarm and ringback and relies on the different visual display indications for differentiation. Sequence R-9 uses a time delay device to silence the ringback audible device after a set time. Sequence R-10 deletes the ringback audible device and uses only the ringback visual displays. Sequences R-9 and R-10 avoid the need for pushbutton operation to silence an audible device when process conditions return to normal. Sequence R-11 uses a common type of flashing for both alarm and ringback and relies on the different audible devices for differentiation. See 4.5, Option Number Designations.

Sequence R retains both momentary and maintained alarms in the alarm state until acknowledged. Sequence R-12 causes momentary alarms to go to the ringback sequence state as soon as process conditions return to normal. New momentary alarms are evident sooner, but may be confused with existing alarms in the ringback state.

Sequence R-1-2-9, shown as an example of the use of option number designations in Figure 5, of this Standard, includes a silence pushbutton and interlock to allow new alarms to be reviewed or logged while flashing after the audible signal has been silenced and to require operation of the silence pushbutton first. In addition, the ringback audible device is silenced after a set time to retain a ringback signal while avoiding the need for pushbutton operation to silence the ringback audible device when process conditions return to normal.

A.3.6 First out

First out annunciators are used to indicate which one of a group of alarm points operated first. To accomplish this, the visual display indication for the alarm point that operates first must be different from the visual display indication for subsequent alarm points in that group. Only one first out alarm indication can exist in any one first out group.

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When first out annunciators are used primarily to identify the first alarm, a flashing visual display can be used to indicate the first alarm, and visual displays without flashing can be used to indicate subsequent alarms. With this approach, the visual display indication for subsequent alarms does not differentiate between new and acknowledged alarms. Two methods using this approach are in common use. (1) First out designation F1 designates a first out sequence with no subsequent alarm state. Subsequent alarms appear in the acknowledged state. Subsequent visual displays do not flash. The audible device does not operate when subsequent alarms occur, unless still operating from the first alarm. The first out indication is reset by the acknowledge pushbutton. It should be noted that subsequent alarms do not lock in when sequence F1A is used. (2) First out designation F2 designates a first out sequence with no flashing for subsequent alarms. The audible device operates when subsequent alarms occur. The first out indication is reset by the acknowledge pushbutton.

To allow the first out visual display indication to be reviewed or logged after silencing the audible device when using first out designations F1 and F2, a separate silence pushbutton should be used in addition to the other annunciator pushbuttons. The silence pushbutton feature is designated by option number 1. — see 4.5, Option Number Designations.

When use of the annunciator requires differentiation between new and acknowledged subsequent alarms, the first out sequence should include different types of visual display flashing to identify the first alarm while new subsequent alarms are indicated by the usual flashing visual display. One method using this approach is in common use. First out designation F3 designates a first out sequence with first out flashing to identify new and acknowledged first alarms and a first out reset pushbutton to reset the first out indication, whether the process condition was returned to normal or not. If desired, an interlock can be provided to require operation of the acknowledge pushbutton before the first out indication can be reset by the first out reset pushbutton by use of option number 3—see 4.5, Option Number Designations.

After the first out indication is reset, that alarm point indicates the process condition in the same manner as subsequent alarms. The next alarm point to operate will display a first out indication.

First out sequences can be automatic reset or manual reset or can provide ringback indication when alarms return to normal. First out sequences are designated by a combination of the first out designation, the basic sequence letter designation, and any required option numbers. Refer to Figure 1, in this Standard for a first out sequence designation example.

Sequence designations for a range of first out sequences are listed in 4.6, First Out Designations. Sequence diagrams and sequence tables for three of the common first out sequences are shown in Figures 6, 7, and 8 of this Standard. Because of the complex nature of first out annunciator sequences, use of those sequences that are readily available should be considered when making a selection.

When annunciators include both first out alarm points and alarm points without the first out feature, the first out windows and alarm modules are usually located together for easy recognition and to facilitate first out logic bus interconnections. Colored window bezels, windows, or lamps may be used to identify first out alarm points. Several separate first out groups can be created by using several first out logic buses.

Annunciator first out windows may not be needed if a recording annunciator or computer printer is provided for sequence of events analysis.

A.3.7 Test

Annunciator test pushbuttons initiate an annunciator sequence to reveal lamp or circuit failures. Most annunciators include a test pushbutton and operational test of the sequence, visual display lamps, audible devices, and pushbuttons. The sequences in this Standard include operational test as a standard feature.

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Operation of the test pushbutton simulates simultaneous abnormal process conditions on all related alarm points. Release of the test pushbutton simulates return to normal. Operation of the other annunciator pushbuttons to complete the sequence and observation of the operation of the visual displays and audible devices can reveal lamp or circuit failures. Alarm points in the acknowledged state as a result of actual abnormal process conditions usually remain in the acknowledged state during test and will be in the acknowledged state after the test sequence.

Since the test signal usually operates the sequence logic circuit in the alarm module, annunciator input auxiliary circuits such as reflash or time delay circuits and some auxiliary output circuits may not operate during the test sequence. In some cases, operation of auxiliary outputs during the test sequence may not be desirable since false alarm signals would be transmitted to the connected alarm, recording, or control circuits.

Some annunciator sequences make use of visual display indications that are dim in place of off to reveal lamp failures in addition to an operational test sequence. The dim lamp monitor feature is designated by option number 13—see 4.5, Option Number Designations.

Test pushbuttons used with some electromechanical relay logic annunciators test only the visual display lamps. Operation of the test pushbutton turns all related visual display lamps on simultaneously. The lamp test feature is designated by option number 14—see 4.5, Option Number Designations.

A.4 Arrangement

A.4.1 Integral logic annunciators

Integral logic annunciators include visual displays and sequence logic circuits in one assembly as illustrated in Figure A.2. Plug-in alarm modules contain the sequence logic circuits. The visual display lamps and windows may or may not be a part of the plug-in alarm modules. Terminal blocks are provided for the field contact, pushbutton, audible device, and power source wiring.

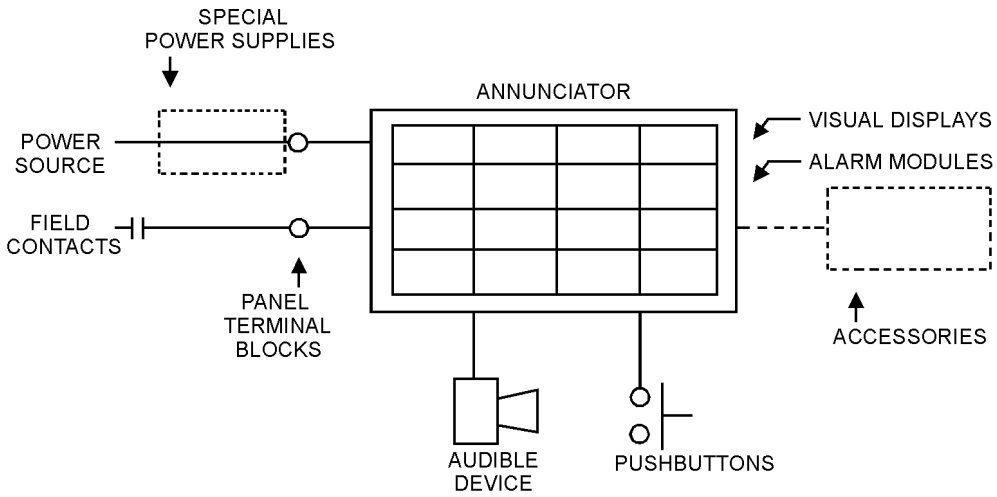


Figure A.2 — Integral logic annunciator arrangement

When large annunciator windows are subdivided to form two, three, or four smaller windows, a single plug-in alarm module may serve each group of alarm points.

Common flashers, audible device drivers, and required power supplies are usually located within integral logic annunciators. Small pushbuttons and audible devices can be mounted in window positions when preferred and when annunciators are easily accessible.

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Since the space in integral logic annunciator cabinets is limited, it may be necessary to locate special power supplies and accessories such as auxiliary output, reflash, ground detector, and power failure detector components in separate cabinets. The separate cabinets are usually designed for surface mounting.

Integral logic annunciators should be located so that alarm module replacement and other required maintenance can be done without excessive interference with nearby activities.

Panel wiring is often used to connect from the annunciator terminal blocks to larger panel terminal blocks where the field wiring is terminated.

A.4.2 Remote logic annunciators

Remote logic annunciators locate visual displays and sequence logic circuits in separate assemblies as illustrated in Figure A.3. Lamp cabinets contain the visual displays and the necessary terminal blocks or plug connectors. The plug-in alarm modules, accessory components, and special power supplies can be supplied in separate assemblies for rack or cabinet mounting by the user, or else complete logic cabinets can be supplied by the manufacturer. Pushbuttons and audible devices are usually located near the lamp cabinets.

Logic cabinets are usually arranged to suit the requirements of a particular annunciator specification. The arrangement can include space for power supplies and accessories such as auxiliary output, reflash, ground detector, and power failure detector components. A ground bus may be required and electrical receptacles or lights may be desirable for use during maintenance. Large terminal blocks and adequate space for field wiring and terminations can be provided.

When required, terminal blocks can be specified to be separated from the logic circuits to minimize circuit damage during installation or to allow installation of terminal blocks and field wiring before the logic circuits are installed.

To reduce congestion, logic cabinets are often located such that field contact wiring avoids control panel wiring areas. Special cables with large numbers of small conductors are usually used to connect between the logic cabinets and the lamp cabinets. These cables often connect directly to the lamp cabinets so that panel wiring and terminal blocks are not required. Installation time may be reduced if prefabricated lamp cabinet cables with plug connectors are provided by the manufacturer. Care must be taken when specifying the length of prefabricated cables with plug connectors on both ends. In some cases, plug connectors on one end of each cable are installed in the field after the cables are cut to the correct length.

The higher cost of remote logic annunciators is often justified for large or complex annunciator systems. Installation and maintenance costs are reduced since the logic cabinets can contain special power supplies and accessory components, field contact wiring terminates away from control panels, special requirements for field wiring space and terminal blocks can be met, and lamp replacement is the only annunciator maintenance required in the control panel area.

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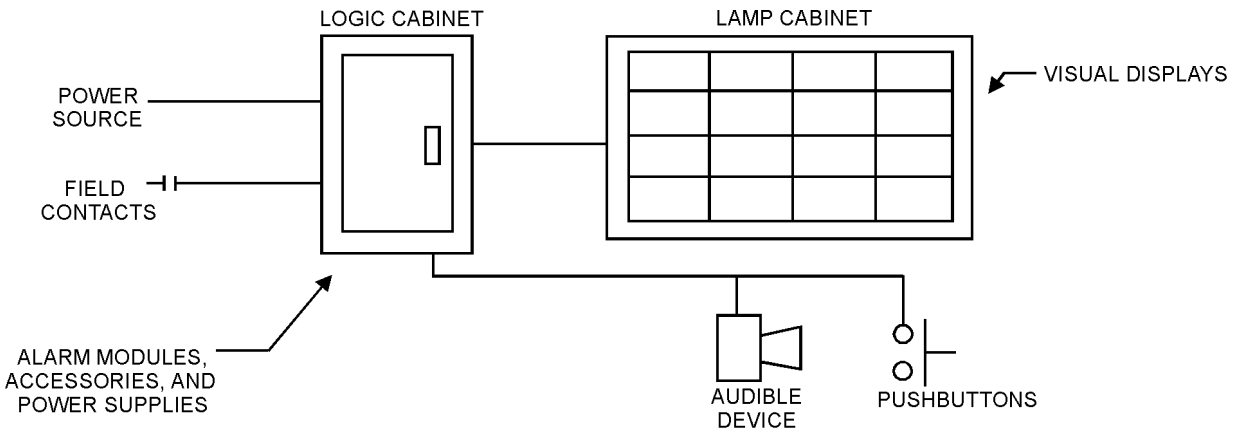


Figure A.3 — Remote logic annunciator arrangement

A.5 Special annunciators

A.5.1 General

The more commonly used annunciators include groups of illuminated visual displays and sequence logic circuits that are dedicated for use with individual alarm points. The following special types of annunciators and visual displays should also be considered when their use can improve the annunciation function. The terminology, sequence designations, and sequence presentation in this Standard should be used wherever practicable when specifying these special annunciators.

A.5.2 Drop type annunciators

Drop type annunciators make use of inexpensive electromechanical devices that release labeled cards to fall to a position where the labels can be seen. A separate device operates the audible device. The audible device and drops can be reset independently by mechanical levers or electrical pushbuttons. Auxiliary output contacts are available.

Drop type annunciators are sometimes used at unattended locations to transmit general trouble alarms to central control room annunciators and to indicate the specific abnormal condition to maintenance personnel.

A.5.3 Individual visual displays

Control panel arrangements sometimes require that annunciator visual displays be located adjacent to related control and indication devices. Individual light boxes or pilot lights suitable for location on panels can be used as visual displays with remote logic annunciators. Some individual light boxes contain the sequence logic and can operate as individual annunciators.

A.5.4 Graphic displays

Graphic displays with colored shapes to represent process equipment and interconnections are often used to enable a more clear understanding of the process operation. Individual visual displays can be included as a part of the graphic display to show the location of alarms in the process. The individual visual displays can be a part of remote logic annunciators or they can operate in parallel with the visual displays of integral logic annunciators.

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A.5.5 Recording annunciators

Recording annunciators, often called sequence of events recorders, provide a printed record of the alarm point identification and the date and time that alarms occur and return to normal. The printed identification to identify the alarms can be coded alarm point numbers or short worded alarm messages.

Recording annunciator circuits usually scan field contacts rapidly, store information when alarms occur in rapid succession, and print the information in chronological sequence. The time can be recorded to as close as the nearest millisecond to allow accurate analysis of the sequence of events. The printers are usually separate from the circuit cabinets.

Recording annunciators are sometimes used in parallel with illuminated visual display annunciators to record the operation of important alarm points. Since they cannot be read from a distance, recording annunciators can be used to both indicate and record alarms only when there is no need to react to alarms quickly or from different control locations. An alarm summary can be printed when required to indicate all points in alarm at that time.

Printers and circuit cabinets can be provided just for recording alarms, or they can be part of larger computer systems that also perform other functions.

A.5.6 Electronic Screen Displays

Electronic Screen Displays can be used as alarm visual displays. Alarms can be presented by worded alarm messages or by words or symbols that appear near the related equipment on Electronic Screen Display process graphics. The alarm indications can flash or be colored or both until acknowledged.

The Electronic Screen Displays are usually separate from the sequence logic and Electronic Screen Display control circuit cabinets.

Commonly used annunciators with conventional illuminated visual displays require additional panel space for each additional alarm point. Since Electronic Screen Displays usually indicate only the points that are in alarm, additional alarm points require only additional capacity in the circuit cabinets. On the other hand, alarms must be displayed on the Electronic Screen Displays in sections when a large number of alarms exist at one time since space on an Electronic Screen Display is limited. When a large number of alarms is anticipated, several Electronic Screen Displays should be used. Each display can indicate a selected portion of the alarms.

When Electronic Screen Displays are used, conventional annunciator illuminated visual displays are often installed also to indicate the more important alarms. Since they are dedicated displays and can be located near the related process controls, they direct attention to the trouble area more rapidly than Electronic Screen Display alarm messages.

Electronic Screen Displays and circuit cabinets can be provided for indication of alarms only, or they can be part of larger computer systems that also perform other functions.

A.5.7 Voice annunciators

Voice annunciators use speakers to broadcast verbal alarm messages. The words may be created by electronic synthesis or by playing messages recorded on magnetic tape.

Voice annunciators tend to command immediate response and can be projected over large control and process areas. They can also be used along with visual display annunciators to call attention to the more important alarms.

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A.5.8 Combined annunciation systems

Annunciation systems can combine annunciators of different types in order to best serve each of the required functions. As an example, a large number of alarms could be presented in a compact control room by the combined annunciation system described below.

Important Alarms	Indicated by conventional annunciator visual displays located on the related control panels. Reflash circuits are used to conserve panel space by indicating several related alarms on one visual display.
All Alarms	Indicated on Electronic Screen Displays located in several major control areas. Each alarm indicates only on the related Electronic Screen Display. The Electronic Screen Displays indicate the reflash circuit alarms individually for better alarm definition.
All Alarms and Important Process Status Inputs	Recorded by a printer for later analysis of alarms and process status and the sequence of events.The printer is located away from the control room to minimize noise in the operating area.
Audible Device and Pushbuttons	Designed to allow silencing audible devices and acknowledging alarms as a coordinated annunciation system.

A.5.9 Multiplexing

Multiplexing is the use of a single communication channel to transmit many individual signals. Multiplexing systems may include redundant circuits and communication channels for increased reliability. While multiplexing is not inherently related to alarms, it is being used with increasing frequency along with annunciators or as a part of annunciation systems to reduce the amount and the cost of field contact wiring. Multiplexing should be considered when the length of field contact wiring is greater than about 1000 feet (300 m).

A.6 Audible devices

Annunciator audible devices call attention by sound to the occurrence of alarm and return to normal conditions. They are often located inside control panels near the annunciators or lamp cabinets. The sound can usually be heard adequately in the adjacent area. In noisy areas, openings in the panels can be provided to increase the sound or the audible devices can be mounted on top of the panels. Small audible devices can be provided on the front of annunciator cabinets. Audible devices are usually supplied with annunciators to ensure coordination with the audible device driver circuits.

Audible devices can be horns, bells, chimes, buzzers, or speakers operating from adjustable solid state tone generators. The sound can be continuous or intermittent until silenced by manual pushbutton operation or the sound can be silenced automatically after a set time. The sound can be steady or it can fluctuate or warble.

Audible devices in relatively quiet control rooms are often speakers or intermittent chimes that call attention to alarms without undue noise. Audible devices with different sounds or tones may be selected to distinguish between different functions, systems, or levels of alarm importance.

Annunciators in locations that are frequently unattended should silence audible devices automatically after a set time to avoid unnecessary noise and wear of the audible devices. Sequence option numbers 7 and 9 are provided to allow adding the automatic silence feature—see 4.5, Option Number Designations.

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When annunciators are in unattended or noisy areas, attention can be drawn to alarms by large flashing lights mounted on top of the panels. These lights can be connected to the audible device driver circuits.

Audible devices should meet the environmental requirements that apply to other panel mounted devices.

A.7 Pushbuttons

Annunciator pushbuttons are momentary manual switches that cause a change from one annunciator sequence state to another. They are usually located on the panels below the annunciators or lamp cabinets. Pushbuttons can also be provided on the front of annunciator cabinets. Usually one normally open or normally closed pushbutton contact is wired to the annunciator. Pushbuttons can be supplied with annunciators or provided by others.

Miniature or heavy duty pushbuttons can be used, depending on the application and the need to coordinate with other panel devices. Since annunciator pushbuttons may be operated frequently and in a hurry when responding to alarms, they should be selected and located for convenient operation and to minimize the possibility of accidental operation of other nearby pushbuttons.

Pushbuttons should meet the environmental requirements that apply to other panel mounted devices.

An interlock is usually provided to require operation of acknowledge pushbuttons before alarms can be reset. It may also be desirable to require operation of silence and acknowledge pushbuttons or acknowledge and first out reset pushbuttons in sequence to avoid accidental loss of alarm indications. Sequence option numbers 2 and 3 are provided to add these interlocks—see 4.5, Option Number Designations.

When several annunciators or lamp cabinets in a large control room operate together as a system, the pushbuttons should be arranged and connected to allow operation from appropriate locations. As an example, the following arrangement could be used in a control room with separate supervision and control panels.

Silence Pushbuttons	Located on all supervision and control panels and connected to silence the alarm audible device for any alarm in the control room. This avoids continued noise while retaining the flashing visual displays.
Acknowledge Pushbuttons	Located only on control panels where corrective control action can be taken and connected to acknowledge new alarms related to the controls on that panel only. This encourages observation of related indicators and controls.
Reset Pushbuttons	Located on all supervision and control panels and connected to reset any ringback alarm in the control room. Return-to-normal conditions usually do not require control action.
Test Pushbuttons	Located only near annunciators or lamp cabinets and connected to test the related annunciator or lamp cabinet only. This avoids disrupting the entire annunciator system by test at one time.

A.8 Visual displays

A.8.1 General

Annunciator visual displays indicate the sequence state. They are usually enclosures containing lamps behind translucent windows. The windows are labeled to identify the monitored variables. Bullseye pilot lights with round glass lenses and adjacent labeled nameplates can also be used

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as annunciator visual displays. Pilot lights are sometimes better suited for use in adverse environments than are displays with translucent windows.

A.8.2 Windows

The size of annunciator windows has not been standardized among manufacturers. Windows commonly used are in the order of 2 inches (50 mm) high and 3 inches (75 mm) wide. These windows often can be subdivided to form two, three, or four smaller windows to reduce the required panel space when the viewing distance is short or a small amount of lettering is necessary. These alarm points may make use of a common window or alarm module.

Annunciator windows are usually translucent white with black letters. Clear or white lamps are normally used. When it is necessary to uniquely identify some of the alarms in an annunciator, such as more important alarms or alarms related to one process area, it is usually possible to obtain colored window bezels, windows, or lamps. Methods used and colors available vary widely. Lamps are often colored by the addition of colored boots that are slipped over the lamps. The brightness and legibility of displays are reduced when colored windows or lamps are used.

To avoid interfering with more alarms than necessary during relamping, windows or alarm modules usually can be removed individually from the front.

For best visibility, the windows should be approximately perpendicular to the line of sight of viewers. Since annunciators and lamp cabinets are often located above other devices on control panels, some windows are wedge-shaped for better visibility from below. Some control panels are designed with an inclined section at the top for improved visibility. These factors should be considered during the design of control panels.

The size of lettering used on annunciator windows should be selected to be legible from any point in the related operating area. A compromise is required, however, since larger lettering reduces the number of characters available to identify alarms. For uniform appearance, the same lettering size should be used for all windows of a given size in any one control room. Recommended reading distances for different letter sizes are included in ISA Recommended Practice ISA-RP60.6, "Nameplates, Tags, and Labels for Control Centers" (being prepared at the time of publication).

In order to identify the monitored variable completely in the space available, abbreviations are usually required. Abbreviations should be consistent, and periods are usually omitted. For standard abbreviations, refer to American National Standards Institute Standard ANSI Y1.1, "Abbreviations for Use on Drawings and in Text." The lettering on each line is usually centered. To facilitate maintenance, windows can be marked with instrument numbers or small identification numbers to correlate with similar numbers adjacent to alarm modules. Annunciator legend lists should be prepared with care, since the legends will appear exactly as indicated.

The arrangement of information on annunciator windows should be consistent. From top to bottom, the words usually become increasingly specific. For example:

COOLING WATER	PRIMARY PMP A102
HEADER PRESS	BEARING TEMP
LOW	HIGH TRIP

A.8.3 Lamps

Annunciator visual displays usually are incandescent lamps. Two lamps are usually connected in parallel to light the windows more evenly and also to allow visual displays to function even if one lamp has failed. Lamps with bayonet bases are usually used to avoid loss of electrical contact because of vibration.

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To minimize maintenance, annunciator lamps are usually operated below the rated voltage. The particular type of lamp, the rated voltage, and the applied voltage are usually selected by annunciator manufacturers. Lamps with a higher watt rating usually cannot be substituted without damage to annunciator circuits or windows.

Care should be taken so the illumination level in areas where annunciator windows are located is not so high that legibility is reduced below an acceptable level.

A.9 Logic circuits

A.9.1 General

Annunciator logic circuits operate the visual displays and audible devices through a chronological series of states to indicate and call attention to alarm and return to normal conditions. Changes from one sequence state to another are caused by changes in process conditions and also by manual operation of pushbuttons. Annunciators usually operate from field contacts that are located in the devices that sense the process conditions.

Annunciator logic normally uses solid state circuits on plug-in printed circuit boards, although electromechanical relay circuits are also available. The widest variety of annunciator features and accessories is available when solid state circuits are used. Solid state logic circuits usually operate on a low dc voltage, from 5 to 15 volts dc. Electromechanical relay logic circuits usually operate on the annunciator power source voltage.

Even when solid state logic circuits are used, some electromechanical relays may be provided to drive large audible devices and to provide isolation in pushbutton, auxiliary output, and reflash circuits. Miniature relays mounted on printed circuit boards are often used to provide isolation because of small size.

Alarm modules usually include movable jumpers or switches to allow operation with either normally open or normally closed field contacts. Other jumpers or switches may be provided to allow selection of the annunciator sequence.

All of the alarm points of an annunciator usually have the same operating sequence; however, different sequences can be provided for individual alarm points or groups of points in one annunciator.

All of the visual displays of an annunciator or lamp cabinet usually flash in unison to minimize confusion, particularly during test. It is usually not necessary to have all visual displays of a large system flash in unison.

A.9.2 Field contacts

Annunciator field contacts are the electrical output contacts of the devices sensing the process conditions. In relation to annunciators, field contacts are either normally open and close to alarm or they are normally closed and open to alarm. This contact operation description is usually not adequate when specifying the devices that sense the process conditions. It is usually necessary to specify that the contacts of these devices close or open when the process conditions increase or decrease. Field contacts may be either electromechanical or solid state.

Movable jumpers or switches are usually provided as a part of alarm modules to allow sequence logic circuits to operate from either normally open or normally closed field contacts. To avoid confusion during maintenance, however, all annunciator field contacts used with one annunciator or in one installation are usually selected to be of the same type.

Many annunciators are used with normally open field contacts and many are used with normally closed contacts. Some applications and industries are fairly consistent in this selection, while others are not consistent. The selection is most often based on past practice, but differences in

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annunciator operation when failures occur and during maintenance can be factors. Failures in the mechanical or electrical devices sensing the process conditions affect both normally open and normally closed field contacts. Open circuits in the field contact wiring prevent alarms from normally open contacts, but cause false alarms from normally closed contacts. Short circuits in the field wiring cause just the opposite results. Disconnecting field contact wiring briefly to locate grounded wires temporarily prevents alarms from normally open contacts, but causes false alarms from normally closed contacts. Spare alarm points with normally closed field contacts require shorting jumpers on the field contact terminal blocks or some other action to avoid the continuous display of alarms.

The field contact voltage is the voltage applied to the field contacts by the annunciator circuits. It is usually from 12 to 125 volts and can be either ac or dc. Annunciators with normally open field contacts often use from 48 to 125 volts for more reliable operation when electromechanical field contacts are contaminated. Lower voltages may be used with normally closed field contacts where contamination cannot prevent an alarm and also to be compatible with other low voltage instrument and control circuits. A single voltage is used when both field contact types are used with one annunciator.

The annunciator response time is the time period between the operation of the field contacts and initiation of the alarm state. Momentary field contact operations with a duration of less than the response time do not initiate alarms. Annunciator response times are in the order of 5 to 50 milliseconds, depending on the manufacturer. Electrical transients induced in the field contact wiring may cause false annunciator operation if the response time is not long enough. Because of the filtering provided by the response time, annunciators usually do not require shielded field contact wiring.

Longer response times or special time delay circuits can usually be provided when required to prevent false operation caused by extraneous circuit pulses or to avoid nuisance alarms caused by momentary operation of field contacts.

The steady state current that passes through field contacts from annunciators is usually in the order of 1 to 10 milliamperes, depending on the manufacturer. This current is necessary to minimize false operation caused by electrical transients and to provide enough arcing to assist in cleaning electromechanical field contacts. This current is usually sufficient for proper operation of solid state field contacts.

The maximum length of field contact wiring that can be used with annunciators depends on the conductor dc resistance or ac impedance, leakage resistance, the field contact voltage, and the annunciator input circuit design. Special design may be required when field contact wiring involves unusually small conductors or long lengths.

A.9.3 Shared alarm points

In many cases, annunciator alarm points are connected to only one field contact. Sometimes alarm points operate from two or more contacts that are wired in a series or parallel arrangement to develop the required alarm logic or to avoid nuisance alarms during certain operating or maintenance conditions. These series or parallel contact arrangements operate together as a single field contact in relation to the annunciator alarm point.

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In order to conserve space on control panels, it may be desirable to use one annunciator visual display for more than one related monitored variable. This can be done by operating alarm points from two or more normally open field contacts in parallel or normally closed field contacts in series so any field contact operation causes an alarm. When this is done, that alarm point alarms when the first abnormal process condition occurs, and can return to normal only after all process conditions have returned to normal. That alarm point cannot alarm subsequent abnormal process conditions that occur while the first abnormal process condition still exists. This method is convenient and economical because it does not require special annunciator circuits, but it should be used only when knowledge of the occurrence of subsequent abnormal process conditions is not required.

The loss of information that occurs when two or more field contacts are connected in parallel or series does not occur if reflash circuits are added to the annunciator. Reflash circuits allow two or more abnormal process conditions to initiate or reinitiate the alarm state of one annunciator alarm point at any time. Reflash alarm points cannot return to normal until all related process conditions return to normal.

Two or more field contacts are wired independently to reflash circuits. The reflash circuit outputs are connected to sequence logic circuits. The annunciator sequence operates in the usual manner when the first abnormal process condition occurs. When subsequent abnormal process conditions occur while previous abnormal process conditions still exist, the reflash circuit causes the sequence to go to the alarm state again. The sequence cannot return to the normal state until all related process conditions return to normal. Reflash circuits allow annunciators to alarm the first and all subsequent abnormal process conditions, but the identity of the individual monitored variables must be determined from nearby instruments, field instruments, or observation of process conditions.

The number of individual reflash circuits on one reflash module is determined by the manufacturer. Reflash circuit outputs are interconnected to accommodate the required number of field contacts and then wired to alarm modules. Unused individual reflash circuits on some reflash modules may not be available for use with another group, depending on the module internal connections. In some cases, annunciator circuit design limits the number of reflash circuits that can be in any one reflash group.

Reflash modules are usually located together and not intermixed with alarm modules. This allows flexibility as the annunciator requirements develop and change.

A.9.4 Analog input alarm points

While annunciators usually operate from field contacts, they can operate from analog input signals also. Analog input points are for use with analog monitored variable signals. Additional logic circuits initiate alarms when the analog signals are above or below set points.

Analog input points can operate from a variety of dc currents and voltages. In some cases thermocouples, resistance temperature detectors, and ac currents and voltages can be used also. When required, terminal block points should be specified for grounding the shields of analog input point field wiring.

Solid state circuits compare the analog input signals against set points established on the printed circuit boards by adjustable potentiometer. Movable jumpers or switches can be set to determine whether alarms occur when the signals are above or below the set points. The analog signal dead band is usually adjustable.

Separate analog input modules have output contacts for connection to alarm modules. In some cases, the analog input circuits and sequence logic circuits are included in one module.

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A.9.5 Auxiliary outputs

Annunciator auxiliary outputs are operated by single alarm points or groups of points for use with remote devices. Auxiliary outputs may be contact or voltage signals that are connected to remote alarm, recording, or control devices. Auxiliary output contacts may be electromechanical or solid state. These contacts are usually isolated electrically from the annunciator and powered from the remote device. Auxiliary output contacts are usually available that are normally open and closed to alarm or are normally closed and open to alarm.

The four common types of auxiliary output and the operation of a graphic display in parallel with annunciator visual displays are shown functionally in Figure A.4. The type selected for a particular application depends on the required operation. The operation of auxiliary outputs should be added to sequence diagrams and sequence tables or the operation should be defined by notes.

Field contact follower auxiliary outputs operate while individual field contacts indicate abnormal process conditions. The outputs usually do not operate during annunciator test. When field contact follower operation is required, it may be possible to avoid auxiliary outputs by operating annunciators in parallel with other devices using the same field contacts. The possibility and any technical problems should be discussed with the manufacturers of the annunciator and other device.

Lamp follower auxiliary outputs operate while individual visual display lamps indicate alarm, silenced, or acknowledged states. The outputs do not change when visual display lamps flash. The outputs usually operate during operational test, but not during lamp test.

Audible device follower auxiliary outputs operate while the common alarm audible devices operate. The outputs usually operate during operational test, but not during lamp test.

Reflash auxiliary outputs operate when any one of a group of alarm points indicates an abnormal process condition. The outputs usually return to normal briefly when each alarm point changes to an abnormal process condition the outputs return to normal when all alarm points in the group indicate normal process conditions. The outputs usually operate during annunciator test.

Consideration should be given to the operation of auxiliary outputs during annunciator test. In some cases operation of auxiliary outputs can test the remote devices also; however, in other cases operation during annunciator test would be undesirable or unacceptable.

When auxiliary outputs are connected to recording devices for analysis of sequence of events, the response time of the annunciator input and auxiliary output circuits should be considered. This delay may change the recorded sequence when compared to signals that do not pass through the annunciator.

Some installations use annunciator auxiliary outputs for control and protection interlocks and others use separate field contacts. The use of auxiliary outputs is often convenient, but separate field contacts may provide improved operating reliability and avoid unexpected control actions during annunciator maintenance.

Auxiliary outputs are often used to retransmit alarms from local annunciators to annunciators in central control rooms. Local annunciator auxiliary outputs can retransmit the operation of an entire local annunciator, groups of alarm points, or individual alarm points. Outputs from local annunciator power failure detectors are often transmitted also. When a group of local annunciator alarm points is retransmitted to one alarm point in the central control room, the central annunciator indicates that a local alarm has occurred but not which alarm. The following methods are commonly used to retransmit the operation of local annunciators.

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Audible Device Follower Auxiliary Outputs	The central control room annunciator operates when local alarms occur and can return to normal after they are acknowledged at local annunciators. The central control room annunciator does not indicate when local alarms return to normal or that local alarms continue to exist. Operators acknowledging alarms at local annunciators are responsible for correcting all local alarm conditions. This method is convenient, economical, and can be added easily to existing local annunciators.
Reflash Auxiliary Outputs	The central control room annunciator operates when each local alarm occurs and can return to normal after all local alarms return to normal. The central control room annunciator does not indicate when individual local alarms return to normal. This method requires reflash auxiliary outputs in local annunciators and field wires to the control room annunciator from each local annunciator.
Field Contact or Lamp	Reflash alarm points in the central control room annunciator are Follower Auxiliary Outputs used to indicate when each local alarm occurs and when all local alarms return to normal. The central control room annunciator does not indicate when individual local alarms return to normal. This method requires field contact or lamp follower auxiliary outputs in local annunciators, reflash alarm points in the central room annunciator, and field wires to the control room annunciator for each local annunciator alarm point.

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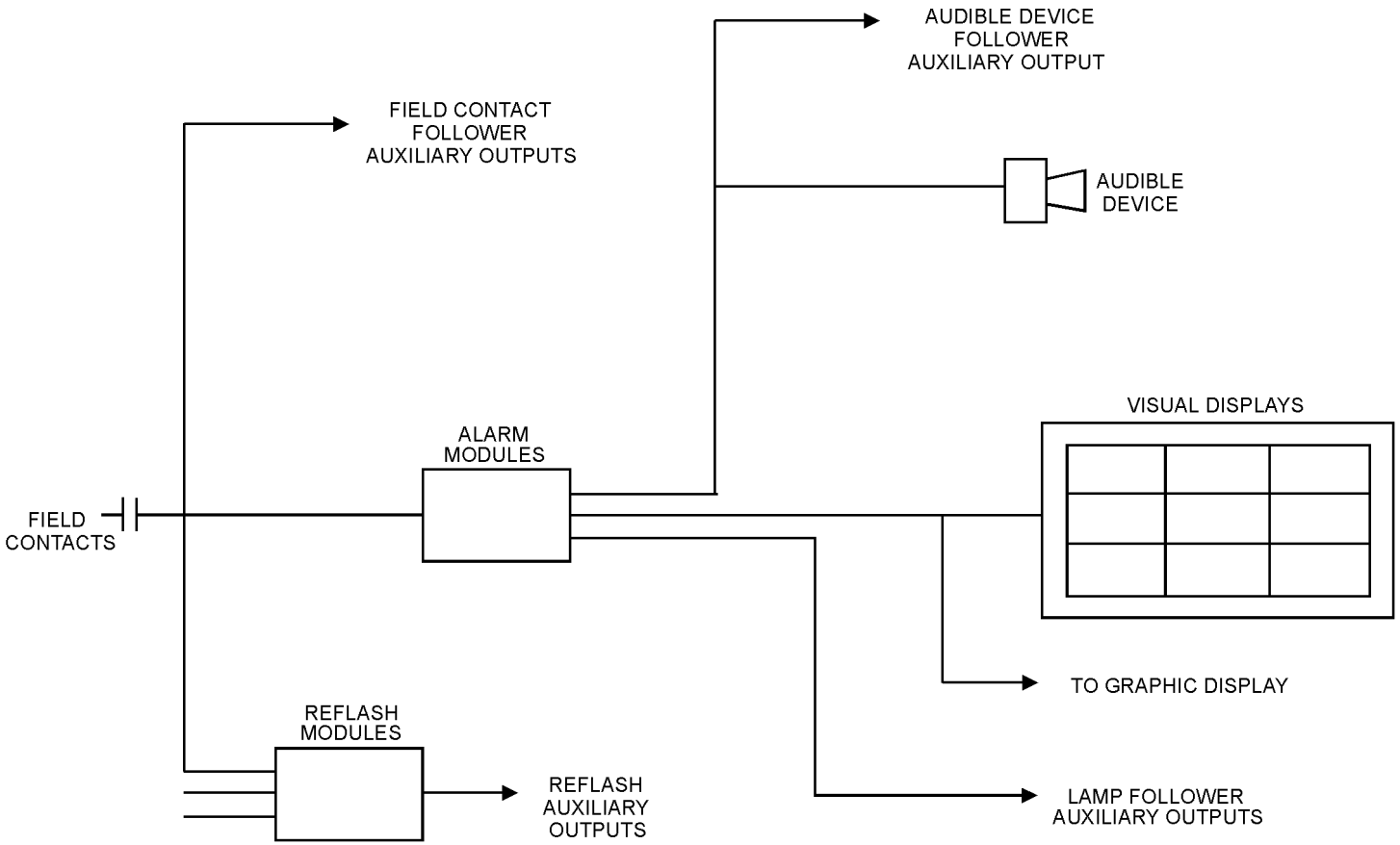


Figure A.4 — Auxiliary output functional diagram

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A.10 Hazardous atmosphere location annunciators

A.10.1 General

Special precautions are required when electrical equipment and circuits are installed in locations where fires or explosion hazards exist or may exist. Hazardous atmosphere locations are classified by class, division, and group. Class I locations are those in which flammable gases or vapors are or may be present in quantities sufficient to produce ignitable or explosive mixtures. Class II locations are those that are hazardous because of the presence of combustible dust. Class III locations are those that are hazardous because of the presence of easily ignitable fibers. Class I, Division 1, includes locations in which the gas or vapor hazard exists continuously or intermittently. Class I, Division 2, includes locations in which hazardous materials are normally confined but can escape when failures or abnormal conditions occur. Specific hazardous atmospheres are identified by group letters.

Special circuit or enclosure design may be avoided in Class I, Division 2, locations when current interrupting contacts are hermetically sealed against the entrance of gases and vapors.

For detailed application and installation information, refer to (1) ISA Recommended Practice ISA-RP12.1, "Electrical Instruments in Hazardous Atmospheres," and (2) National Fire Protection Association Standard NFPA 70, "National Electrical Code," Articles 500-503.

Annunciators for use in hazardous atmosphere locations may require certification to meet the requirements of specific codes or agencies.

A.10.2 Intrinsically safe annunciator systems

Intrinsically safe annunciator systems are designed and installed so that field contacts and field contact wiring are incapable of releasing sufficient electrical or thermal energy under normal or abnormal conditions to cause ignition of specific hazardous gas or vapor atmospheric mixtures in the most easily ignited concentrations. Field contacts and field contact wiring connected to intrinsically safe annunciators can be installed in many Class I, Division 1 and Division 2, hazardous locations without the use of explosion proof enclosures.

Intrinsically safe annunciator systems limit the release of energy from field contacts and field contact wiring under both normal and abnormal conditions by special power supply design, low field contact voltage, current limiting resistors, wiring barriers or separation from nonintrinsically safe circuits, and limited field contact wiring capacitance and inductance. Abnormal conditions include any two independent device or circuit failures in combination.

The annunciator cabinets, power supplies, pushbuttons, and audible devices of intrinsically safe annunciator systems are not intrinsically safe themselves. They must be installed in nonhazardous areas or must be designed for the specific hazardous atmosphere involved. Refer to A.10.3, Nonincendive Annunciators; A.10.4, Purged Cabinets; and A.10.5, Explosion Proof Cabinets.

In some applications, increased flexibility is obtained by the use of special relays and power supplies in the nonhazardous annunciator area to provide intrinsically safe barriers between the annunciator and intrinsically safe field contacts and field contact wiring. These barriers can be supplied with annunciators.

For detailed application and installation information, refer to (1) information from annunciator manufacturers, (2) ISA Recommended Practice ISA-RP12.6, "Installation of Intrinsically Safe Instrument Systems in Class I Hazardous Locations," and (3) National Fire Protection Association Standard NFPA 493, "Intrinsically Safe Apparatus in Class I, II, and III, Division 1, Hazardous Locations."

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A.10.3 Nonincendive annunciators

Nonincendive annunciators are designed so that annunciator cabinets, power supplies, pushbuttons, and audible devices are incapable of releasing sufficient electrical or thermal energy under normal operating conditions to cause ignition of specific hazardous gas or vapor atmospheric mixtures in the most easily ignited concentrations. Nonincendive annunciators can be installed in many Class I, Division 2, hazardous locations without the use of explosion proof cabinets.

The design is not required to consider abnormal conditions such as device or circuit failure. The probability of simultaneous annunciator failure and abnormal conditions causing the escape of hazardous materials is considered negligible. No energy limit is placed on circuits or on contacts, switches, or plugs that are used only during maintenance.

Nonincendive annunciators make use of hermetically sealed contacts where current is interrupted under normal operating conditions. Annunciator cabinet covers that require the use of a tool for removal and approved warning nameplates remind the user to determine that the atmosphere is not hazardous before performing maintenance.

The annunciator field contacts and field contact wiring of nonincendive annunciators are not protected in any way by the nonincendive design. They must be designed and installed for the specific hazardous atmosphere involved. Refer to A.10.2, Intrinsically Safe Annunciator Systems.

For detailed application information, refer to (1) information from annunciator manufacturers, and (2) National Fire Protection Association Standard NFPA 70, "National Electrical Code," Article 501.

A.10.4 Purged cabinets

Annunciators with purged cabinets are designed so that a supply of clean air or inert gas around the electrical devices at sufficient flow reduces hazardous gas or vapors initially present to a safe level and maintains a safe level by positive pressure. Annunciators with purged cabinets can be installed in many Class 1, Division 1 and Division 2, hazardous locations without the use of explosion proof cabinets. Pushbuttons and audible devices that are not in the annunciator cabinet must be designed for the specific hazardous atmosphere involved.

Purging requirements are divided into three classifications. Type X purging reduces the classification of the area within cabinets from Division 1 to nonhazardous, Type Y from Division 1 to Division 2, and Type Z from Division 2 to nonhazardous. Depending on the classification, purged cabinets require special construction, low pressure alarms, nonincendive design, power interlocks, warning nameplates, and special operating procedures.

For detailed application and installation information, refer to (1) information from annunciator manufacturers, (2) ISA Standard ISA-SI2.4, "Instrument Purging for Reduction of Hazardous Area Classification," and (3) National Fire Protection Association Standard NFPA 496, "Purged and Pressurized Enclosures for Electrical Equipment in Hazardous Locations."

A.10.5 Explosion proof cabinets

Annunciators with explosion proof cabinets are designed so that the cabinets are capable of withstanding explosions of a specified gas or vapor that may occur within them while preventing the ignition of specific hazardous gas or vapor atmospheric mixtures surrounding the cabinets. Annunciators with explosion proof cabinets can be installed in many hazardous locations. Pushbuttons and audible devices that are not in the annunciator cabinet must be designed for the specific hazardous atmosphere involved.

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For NEMA enclosure requirements, refer to National Electrical Manufacturers Association Standard NEMA ICS 6, "Enclosures for Industrial Controls and Systems."

A.11 Cabinets

Annunciator cabinets and lamp cabinets are usually flush mounted on steel panels. The cabinet bezels and windows extend about 1 inch (25 mm) in front of the panels. Annunciator cabinets extend further behind the panels than lamp cabinets. Cabinets usually fit into rectangular cutouts in the panels and are held in place by clamps. Access for lamps and alarm modules is usually from the front. Access for wiring is usually from the rear. Openings for ventilation may be provided. Forced ventilation and high temperature alarm contacts are usually not required. Special surface mounted annunciator and lamp cabinets with access from the front are also available.

Logic circuits for use with remote logic annunciators usually can be supplied as open chassis assemblies, enclosed chassis assemblies, or 19 inch (475 mm) rack mount assemblies for mounting in logic cabinets by others. Logic circuits can also be provided in free-standing or surface mounted logic cabinets.

Annunciator cabinets are usually for general purpose indoor installation and are classified as NEMA 1. Special cabinets to meet NEMA 12 indoor driptight and dusttight requirements, and NEMA 4 indoor and outdoor watertight and dusttight requirements may also be available. For NEMA enclosure requirements, refer to National Electrical Manufacturers Association Standard NEMA ICS 6, "Enclosures for Industrial Controls and Systems." For annunciators located in hazardous atmospheres, see A.10, Hazardous Atmosphere Location Annunciators.

A.12 Power supply systems

A.12.1 Power sources

Annunciators are available for use on a wide range of power source voltages, on ac at 50 or 60 hertz, and on dc. The most common power source voltage is 120 volts, 60 hertz, but other ac voltages such as 24, 48 and 230 volts are used also. Annunciators can usually operate over a range of at least 90 to 110 percent of the rated ac voltage.

Many annunciators operate on dc power sources to allow continued operation from batteries in the event ac power is lost. Larger installations usually provide 125 volt dc power for this purpose. Other dc voltages such as 24, 48, and 250 volts are used also. Annunciators can usually operate at the high equalizing charge voltage applied to some batteries during periodic maintenance and the low voltage that may exist during emergency operation. This voltage range is usually from 105 to 140 volts for annunciators designed to operate on 125 volts dc.

Annunciator electrochemical relay logic circuits usually use the power source voltage directly for the logic relays, lamps, and field contacts. Solid state logic circuits usually require power supplies to develop the dc voltages used for logic circuits, lamps, and field contacts.

A.12.2 Power supplies

Annunciator power supplies convert ac or dc power source voltages to one or more dc voltages for use with solid state logic circuits, lamps, and field contacts. Power supplies are usually supplied with annunciators to ensure coordination with the annunciator circuits.

When dc power sources are used, special power supplies may be provided to convert dc power source voltages directly to the required dc annunciator circuit voltages, or separate inverters may be used to invert the dc to ac for connection to conventional ac to dc power supplies.

Power supplies provide electrical isolation between power sources and annunciator circuits. This isolation allows annunciator circuits to operate ungrounded. When accidental circuit grounds

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occur on ungrounded annunciator circuits, ground detectors operate annunciator alarms and the grounds can be located and removed without affecting the annunciator operation or the power source.

Inverters and power supplies used with annunciators should be provided with overload protection and should not require manual actions when voltage is applied or after brief overloads occur.

A.12.3 Backup power

Annunciators are no more reliable than the power on which they operate. The power sources, power supplies, and power distribution systems for annunciators should provide the reliability required by the annunciation function.

A single power supply may provide power for only a few alarm points or for an entire annunciator. An annunciator power source may come from a 120 volt ac lighting panel or from a large battery and charger system. The failure of a single power supply or power source can disable an entire annunciator, or backup power supplies and power sources can be used so that any single failure affects no more than one alarm point. Power supply system designs should suit the power sources available and provide the required annunciator reliability.

While an annunciator powered from a dc power source provides reliable operation, a single failure in the dc distribution panel or annunciator feeder can leave the annunciator without power. In addition, dc power source buses are often ungrounded. When a ground occurs, the load circuits are switched off and on again in turn until the grounded circuit is located. When the annunciator feeder is switched off and on again, the alarm points usually assume a random sequence state and auxiliary outputs may operate incorrectly. It is then necessary for the annunciator pushbuttons to be operated before the correct alarms are displayed again.

When two independent power sources are used to power an important annunciator, the reliability becomes very high. Backup power supplies can also be used so that failure of a single power supply does not affect annunciator operation. When backup power sources or power supplies are used, power failure detectors should be provided to alarm the failure of each power source and power supply.

A.12.4 Power failure detectors

Power failure detectors provide outputs to alarm the loss of voltage on annunciator logic circuit, lamp, and field contact power buses. Power failure detectors cannot alarm on the failed annunciator, so another alarm device must be used.

Local annunciator power failure can be alarmed on central control room annunciators. Separate Electronic Screen Displays or recording annunciators can be used to alarm control room annunciator failure. In some cases, separate annunciator audible and visual devices powered by an independent power source are provided to alarm annunciator power failure. In these cases the independent power source must, in turn, be monitored by the annunciator.

A.12.5 Ground detectors

Annunciator buses are often ungrounded so that accidental circuit grounds can be detected and ground alarms initiated without affecting the operation of annunciators. Grounds on the field contact wiring are the most common since this wiring is often run in areas that have an adverse environment.

Ground detectors are usually connected to alarm the presence of a ground on either conductor of an ungrounded system. The ground should be located and removed before a second ground occurs, since two or more simultaneous grounds can affect annunciator operation.

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Grounds are usually located by disconnecting individual circuits in turn until the ground detector indicates that no ground exists. The ground is on the disconnected circuit.

Annunciator circuits are usually disconnected to check for grounds by briefly disconnecting field contact wires or by briefly removing individual alarm modules. Large annunciators often include power supply system disconnect switches to subdivide the system for more convenient ground isolation and maintenance. Some remote logic annunciators include field contact wiring terminal blocks that include switches or other devices to allow ground isolation without affecting the field wiring terminations.

The location of grounds is made easier if an indication of grounds is within sight of the disconnecting devices. Lights operating from ground detectors can be located in alarm module and terminal block areas for this purpose.

A.13 Factory tests

A.13.1 General

Annunciator factory tests can include prototype, component, module, and final assembly tests. The routine tests performed by the manufacturer are often the only tests performed. In some cases, users specify specific tests to confirm that annunciators will operate as required in special applications.

Annunciators are usually tested using test procedures that apply to industrial electrical and electronic instrumentation and control equipment.

A.13.2 Dielectric strength

Dielectric strength tests, sometimes referred to as high potential or high pot tests, are usually performed on annunciator final assemblies to confirm that insulation and clearances are adequate between circuits and cabinets and also between separate circuits. To avoid damage, solid state circuit modules are usually removed, grounded, or shorted during these tests.

Test voltages range from 1000 to 1500 volts ac and are usually applied for one minute. Dielectric strength standards are included in (1) National Electrical Manufacturers Association Standard NEMA ICS 1, Part ICS 1-109, "Industrial Controls and Systems, Tests and Test Procedures," (2) American National Standards Institute Standard ANSI C39.5, "American National Standard Safety Requirements for Electrical and Electronic Measuring and Controlling Instrumentation," and (3) Underwriters Laboratories Standard UL 508, "Industrial Control Equipment."

A.13.3 Functional

Functional tests are usually performed on annunciator modules and final assemblies to confirm that the operation is as specified. Tests on final assemblies are sometimes referred to as simulation tests.

Functional tests usually include operation in all of the normal operating modes and also in abnormal modes such as high and low power source voltage, grounds, short circuits, and unusual service conditions. Cabinet internal temperatures are often measured to confirm ventilation designs.

Annunciator modules are often subjected to a burn-in period. This reveals modules with weak components that would fail early in the annunciator life. The burn-in period varies widely, depending on the manufacturer or the specification requirements.

Logic circuit test assemblies can usually be provided to facilitate testing individual modules in the field.

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A.13.4 Surge withstand capability

Annunciator solid state logic circuits operate at low power levels. Annunciator circuits are designed to suppress electrical surges and transients that may be induced in field wiring. Tests are usually performed on prototype modules to confirm that expected surges and transients do not cause false operation or damage.

A specific surge withstand capability test is included in Institute of Electrical and Electronics Engineers Standard IEEE 472, "IEEE Guide for Surge Withstand Capability (SWC) Tests."

A.13.5 Radio frequency interference

There is some possibility that false operation of annunciator solid state logic circuits can be caused by operation of nearby portable radio transmitters, especially when cabinet doors are open.

A radio frequency interference test method and a classification of signal field strengths and frequency bands are included in Scientific Apparatus Makers Association Standard SAMA PMC 33.1, "Electromagnetic Susceptibility of Process Control Instrumentation."

A.13.6 Seismic

In some applications, annunciators may be required to operate during and after specified earthquakes, or annunciator cabinets may be required only to remain intact externally so as not to endanger nearby personnel or equipment. When required, seismic tests are usually performed on prototype assemblies.

Seismic test procedures are included in Institute of Electrical and Electronics Engineers Standard IEEE 344, "IEEE Recommended Practices for Seismic Qualification of Class 1E Equipment for Nuclear Power Generating Stations."

A.13.7 Class 1E equipment for nuclear generating stations

Annunciators that perform Class 1E safety related functions in nuclear power generating stations must be qualified to confirm the adequacy of the equipment design under all expected normal and abnormal operation conditions.

Qualification procedures making use of type testing, operating experience, and analysis are included in Institute of Electrical and Electronics Engineers Standard IEEE 323, "IEEE Standard for Qualifying Class 1E Equipment for Nuclear Power Generating Stations."

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Appendix B — Sequence designation conversion

This Appendix is included for informational purposes and is not a part of this Standard.

The sequences in ISA Recommended Practice ISA-RP18.1-1965 that are superseded and the corresponding new sequence designations used in this Standard are listed below.

ISA-RP18.1-1965 Sequences	**ISA-SI8.1 Sequence Designations**
ISA-1	A
ISA-1A	A-5
ISA-1B	A-4
ISA-1C	A-4-5
ISA-1D	A-13
ISA-1E	Special(1)
ISA-1F	Special(1)
ISA-2	Special(1)
ISA-2A	R-8
ISA-2B	R-11
ISA-2C	M
ISA-2D	M-5
ISA-3	Special(2)
ISA-3A	Special(2)
ISA-3B	Special(2)
ISA-4	Special(1)
ISA-4A	F1A

(1) Requires two visual display indication devices.

(2) Requires three visual display indication devices.

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